



Zagazig University

Faculty of Engineering



ECE 321: Electronic Circuits

Lecture (01) & (02) → Introduction, Quick Review

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**ECE 321 Electronic Circuits, ECE dept., Faculty of Engineering, Zagazig University, Fall 2022,
by Dr. Ahmed Wahba**

Outline

- ☐ **Course Information**
- ☐ **Introduction: Semiconductors between past, present, and future**
- ☐ **Quick Review**
 - ☐ **Bipolar Junction Transistor (BJT)**
 - ☐ **Metal Oxide Semiconductor Field Effect Transistor (MOSFET)**
- ☐ **Amplifier Model**

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- ☐ **Amplifier Model**

Course Information

☐ **Lecture Time:**

- ☐ 3 hrs/week, Sunday @ 8:30 to 11:30

☐ **Lecture Place:**

- ☐ Hall 27520

☐ **Tutorial:**

- ☐ 1.5 hrs/week

☐ **Materials:**

- ☐ Lectures, Sheets, References,

☐ **Assessment:**

- ☐ Final Exam (100 pts), Classwork (50 pts) →
[Midterm Exam, Quizzes, Assignments,
Attendance, project,....]

☐ **Instructors:**

- ☐ *Dr. Ahmed Reda & Dr. Ahmed Wahba*
- ☐ Email: awelsayed@zu.edu.eg
- ☐ Office Hours: Wednesday 10:00 to 14:00,
Room 27413

☐ **TAs:**

- ☐ *Eng. Asmaa Gamal & Eng. Mahmoud Ghali*

Course Content (Part I)

☐ **Frequency Response**

- ☐ The DC operating point, DC and AC analysis, Miller effect, low frequency response, high frequency response, total frequency response, device application..etc.

☐ **Operational Amplifier**

- ☐ Modeling of Op-amp, linear and non-linear application, imperfections, circuit implementation....etc.

☐ **Feedback Circuits**

- ☐ Types of feedback, advantages and disadvantages, analysis of feedback circuit.

☐ **signal generator and wave shaping**

- ☐ Multi-vibrator, linear oscillator, analog and digital multi-vibrato

Course Content (Part II)

☐ Quick Review

- ☐ BJT and MOSFET devices (construction, operation, small signal model, DC and AC analysis)

☐ Current Mirrors

- ☐ Different types of current mirrors using BJT or MOSFET (simple, cascode, wilson, ...)

☐ Differential Amplifiers

- ☐ DC operation, large signal operation, common-mode and differential mode response, ...

☐ Operational Amplifiers

- ☐ Different types of op-amps (simple single-stage, cascode, telescopic cascode, folded cascode, and two stage op-amp)

☐ Sinusoidal Oscillation

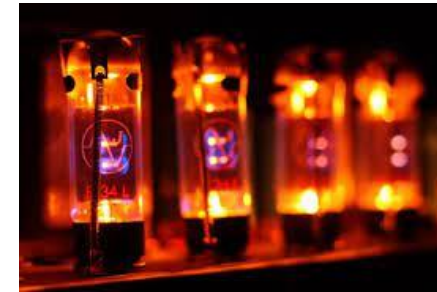
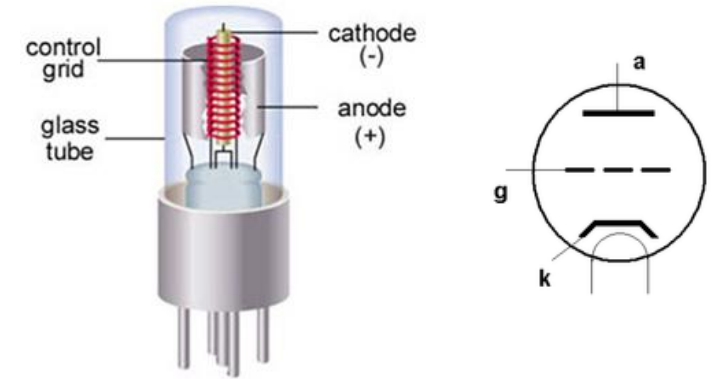
- ☐ Barkhausen Criterion, Wien Bridge, Phase shift Oscillators, LC oscillators

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- ❑ Amplifier Model

Before Transistor: Vacuum tubes

- ❑ **Vacuum tube** was the key component of electronic circuits for the first half of the 20th century (1904 to 1947). It is a device that has **three** terminals (cathode, anode, and control grid)
- ❑ **Operation:** A current is passed through the **cathode** heater, releasing electrons that are attracted to the positively charged **anode**. The flow of electrons is controlled by the **grid voltage** (+Ve voltage accelerates electrons, resulting in high current flow. If a -Ve voltage is applied, the -ve charge will repel electrons, resulting in zero current)
- ❑ **Applications:** used in **radio**, **television**, **radar**, **sound systems**, **telephone networks**, and early digital **computers**. Nowadays, They are rarely used. Check (Why do vacuum tube amplifiers still exist?)
- ❑ **Disadvantages of Vacuum tubes:** large in size, fragile, generally require large voltage to operate, high power consumption, limited lifetime, and expensive.

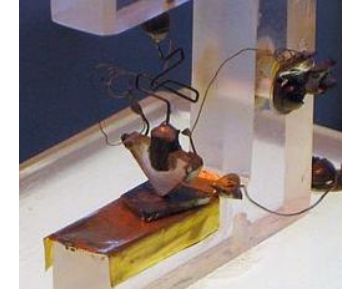


Willsenton
vacuum tube
amplifier

Transistor Invention

- ❑ **In 1947:** Bipolar Junction Transistor (**BJT**) was invented at Bell Labs by **Bardeen, Shockley, and Brattain** (They were awarded the 1956 Nobel Prize in Physics).
- ❑ **In 1959:** The invention of Metal Oxide Semiconductor Field Effect Transistor (**MOSFET**) by **Mohamed Atalla (Egypt)** along with his colleague **Dawon Kahng (Korea)**.
- ❑ Check this documentary about *Prof. Mohamed Atalla* → <https://www.youtube.com/watch?v=RC2tFKNKAto>

Bell Labs 1st transistor



The Nobel Prize in Physics 1956



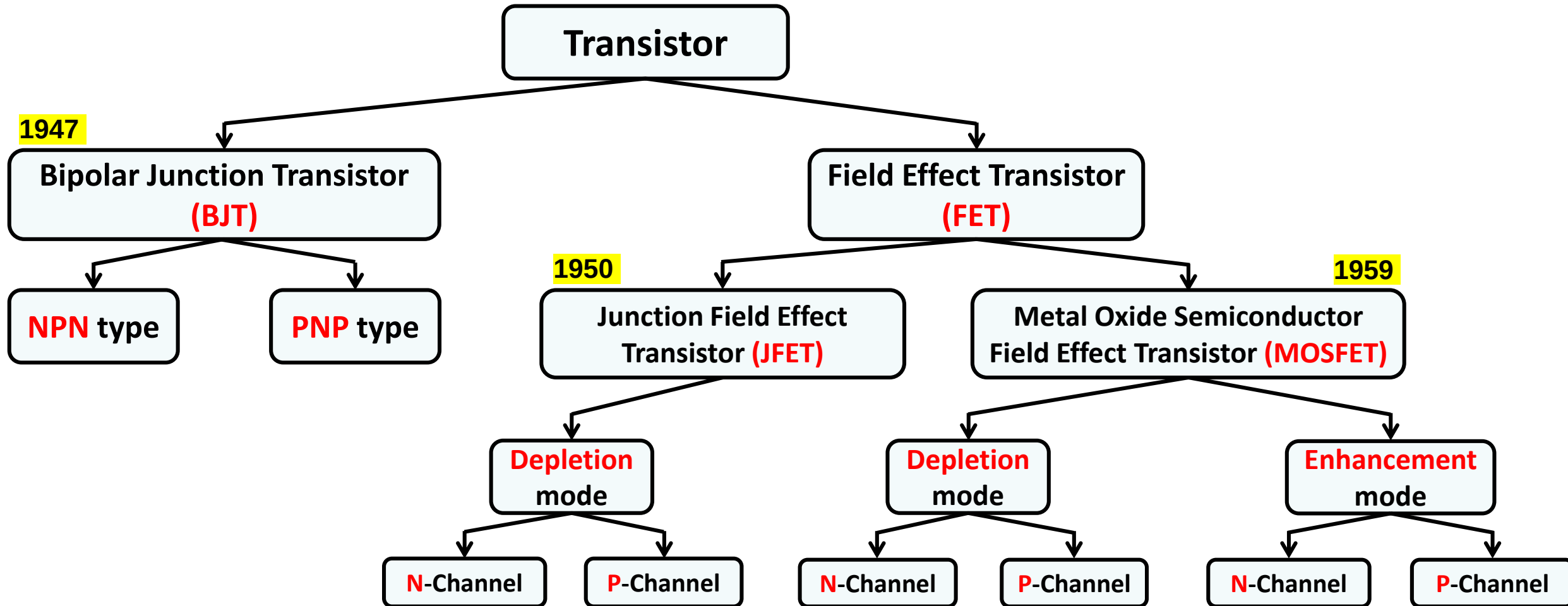
Mohamed Atalla



Dawon Kahng



Transistor Types

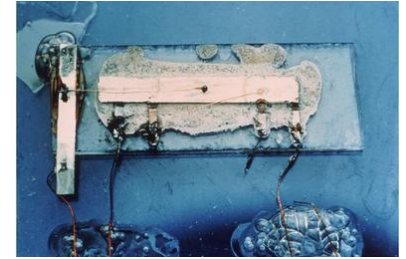


Integrated Circuit (IC) Concept

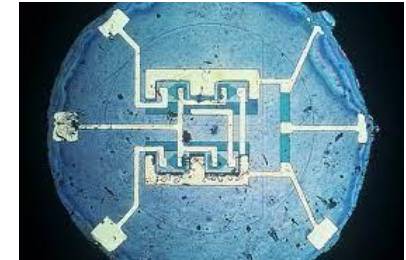
- ❑ In 1959: The concept of integrated circuit (IC) appeared [integrates a large number of electronic components (mostly transistors) on a small piece of semiconductor material].
- ❑ In 1959: Texas Instruments introduced the first IC by Kilby. But it was a hybrid IC (external routing and no isolation) and not suitable for mass production.
- ❑ In 1960: Fairchild introduced the first Monolithic IC by Noyce [Nicknamed: the Mayor of Silicon Valley]. The IC was fabricated using the planar process (Hoerni) and surface passivation process (Mohamed Atalla). The semiconductor industry adopted the same way to manufacture integrated circuits.
- ❑ Today, both Kilby and Noyce are credited with having independently invented the integrated circuit.



Kilby's 1st Hybrid IC



Noyce's 1st Monolithic IC



Jack Kilby



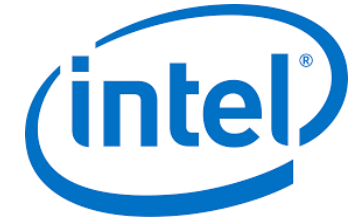
Robert Noyce



Big Companies in Semiconductor Industry

- ❑ **Intel:** founded in 1968 by Noyce, Moore, and Grove in U.S. Grove is credited with having transformed Intel from a manufacturer of memory chips into the world's dominant producer of microprocessors
- ❑ **AMD:** founded in 1969 in U.S. by Sanders along with 7 of his colleagues from Fairchild Semiconductor.
- ❑ **ASML:** founded in 1984 in Netherland. Largest supplier for photolithography machines required to manufacture ICs.
- ❑ **TSMC:** Founded in 1987 by Morris Chang in Taiwan. The world's largest semiconductor foundry.
- ❑ **Other Companies:** (Nvidia, Broadcom, Qualcomm, Texas Instruments, Synopsys, Cadence, ON Semiconductor, IBM, GlobalFoundries,... from U.S.), (Samsung, LG,... from S.Korea), (Huawei, Lenovo, SMIC,... from China), (Sony, Panasonic, Toshiba,... from Japan)

Grove, Noyce, and Moore from left to right

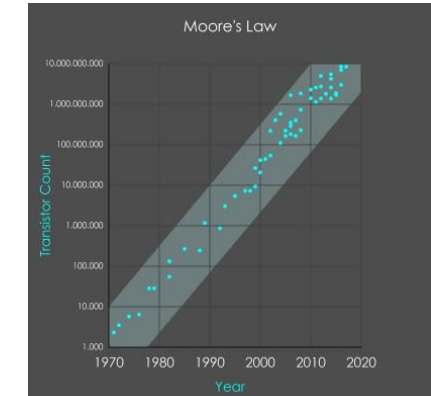
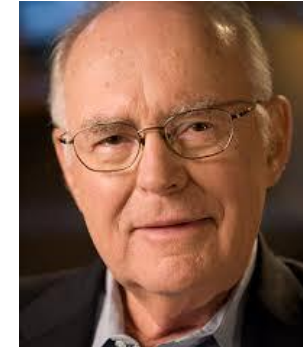


Jerry Sanders



Moore's Law

- ❑ **In 1965:** Moore stated that the number of transistors on a semiconductor chip will **double every year**. Later in **1975**, Moore published an update, revising the **doubling period to every two years**. (Moore's Law)
- ❑ **Discussion:** Does Moore's Law still valid today? What are your future expectations?



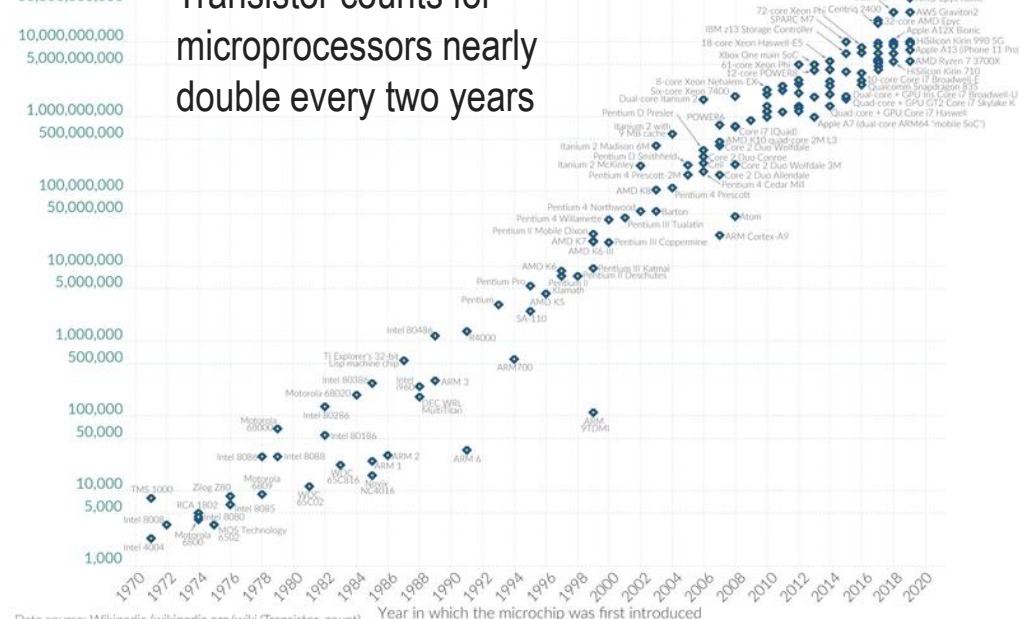
Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

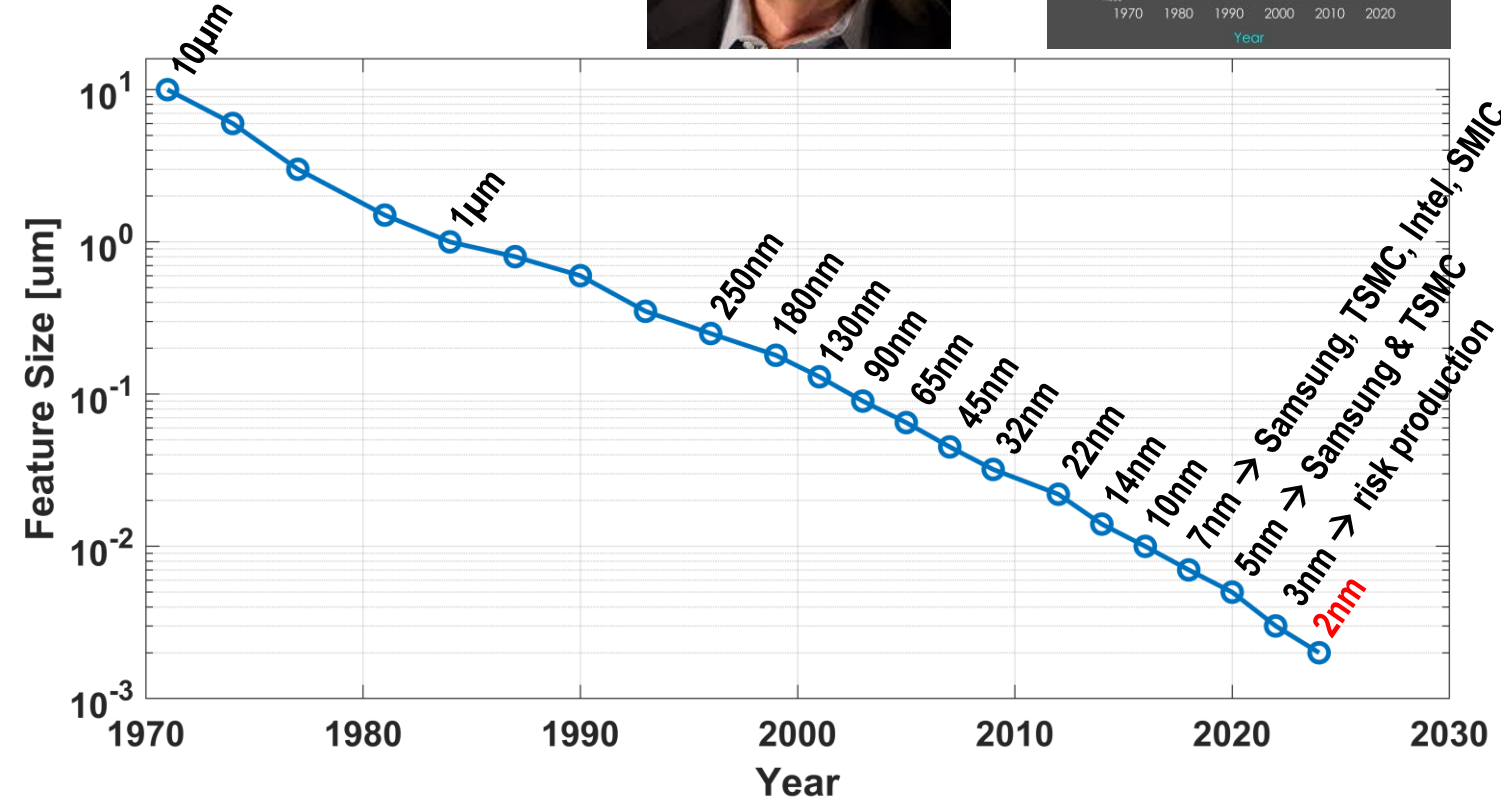
Transistor count

50,000,000,000

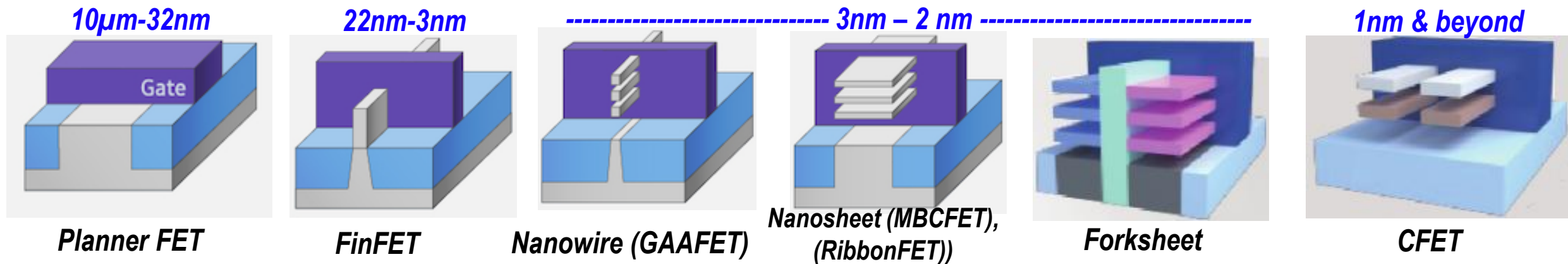
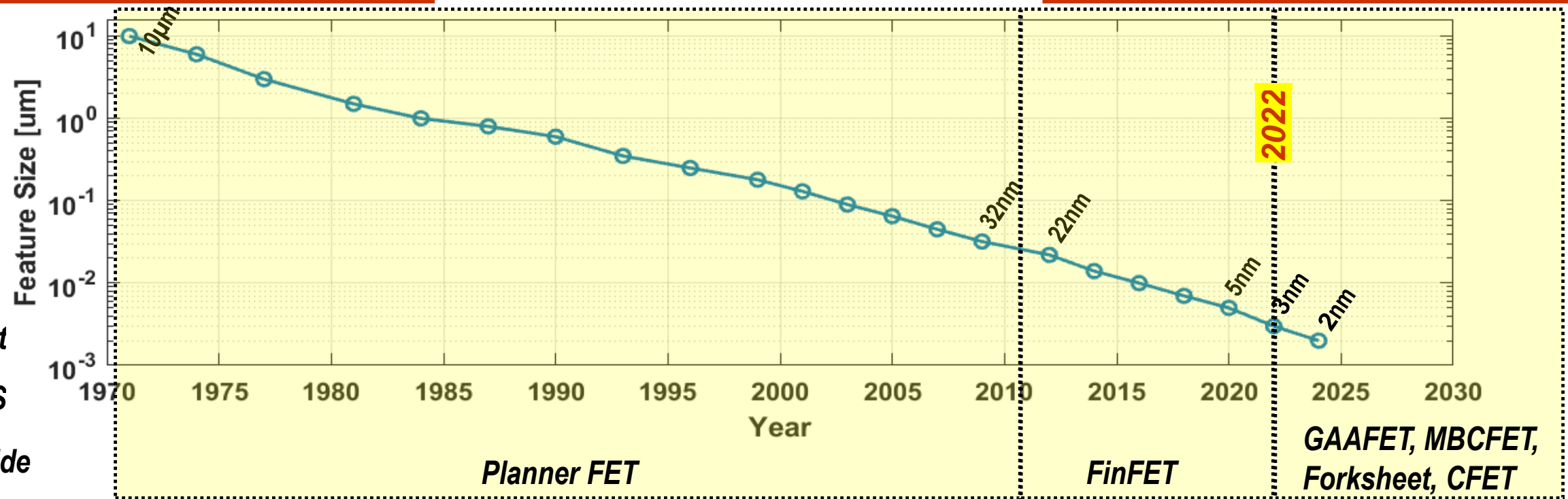
Transistor counts for microprocessors nearly double every two years



Data source: Wikipedia ([wikipedia.org/wiki/Transistor_count](https://www.wikipedia.org/wiki/Transistor_count)) Year in which the microchip was first introduced
OurWorldinData.org - Research and data to make progress against the world's largest problems. Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.



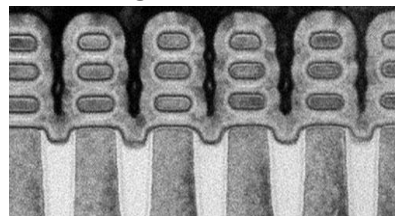
Developments in MOSFET Fabrication



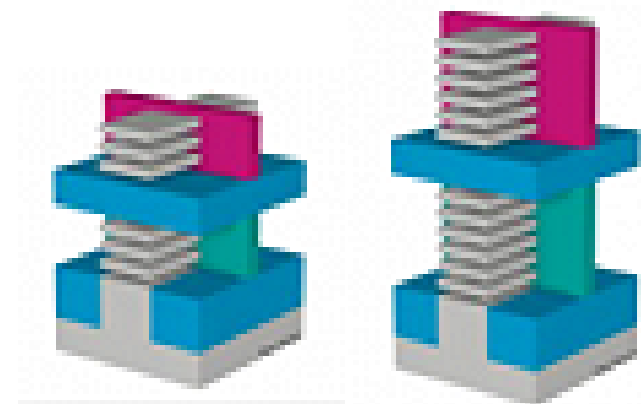
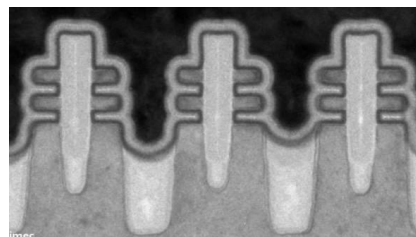
FET Scaling Roadmap

	2022	2024	2026	2028	2030	2032
Process node	3 nm	2 nm	1.4 nm (A14)	1 nm (A10)	0.7 nm (A7)	0.5 nm (A5)
Transistor type	Nanosheets (MBCFET) by Samsung and FinFET by TSMC	Nanosheets	Nanosheets	Forksheet	Forksheet	CFET

Samsung 3 nm nanosheet



Forksheet



CFET: vertical stacking of large number of NFET/PFET

❑ **Si** atom diameter = **0.222 nm** → Atomic or even **single atom** transistor → end of scaling down

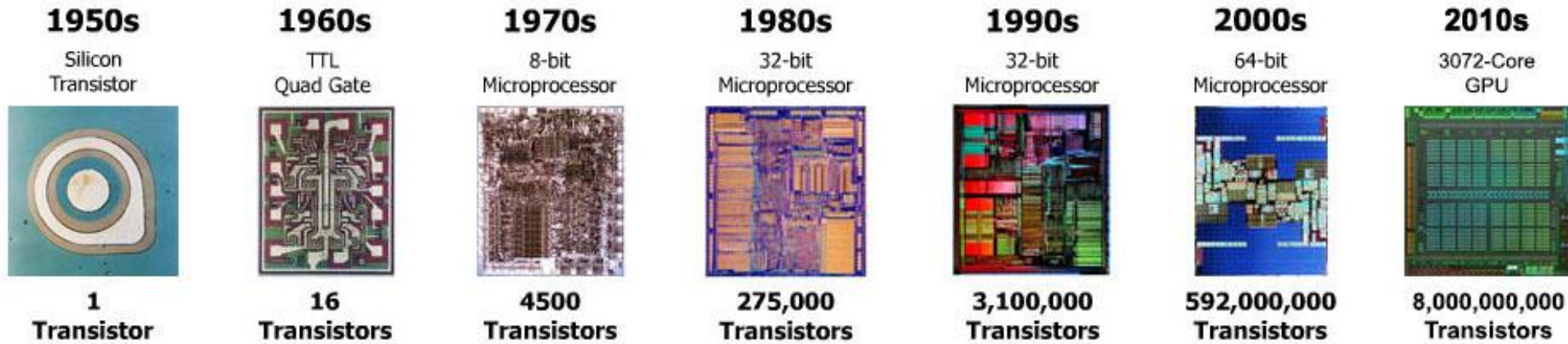
❑ **Challenges:** **Quantum Mechanics** effects start to dominate

❑ **Q: Will Moore's Law die?**

❑ **3D packaging**

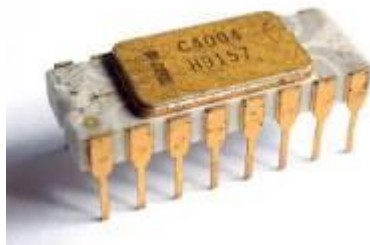
2024 to 2032 roadmap based on → "Imec Presents Sub-1nm Process and Transistor Roadmap Until 2036". Tom's Hardware. 21 May 2022.

Intel Processors

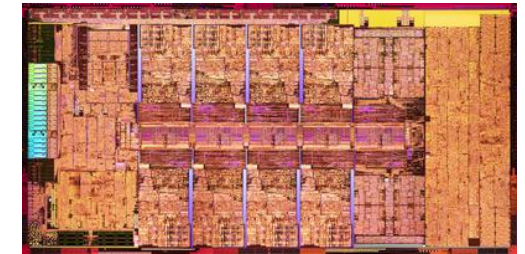


Timeline of microprocessors

Intel Processors between past and present:



51 years of development



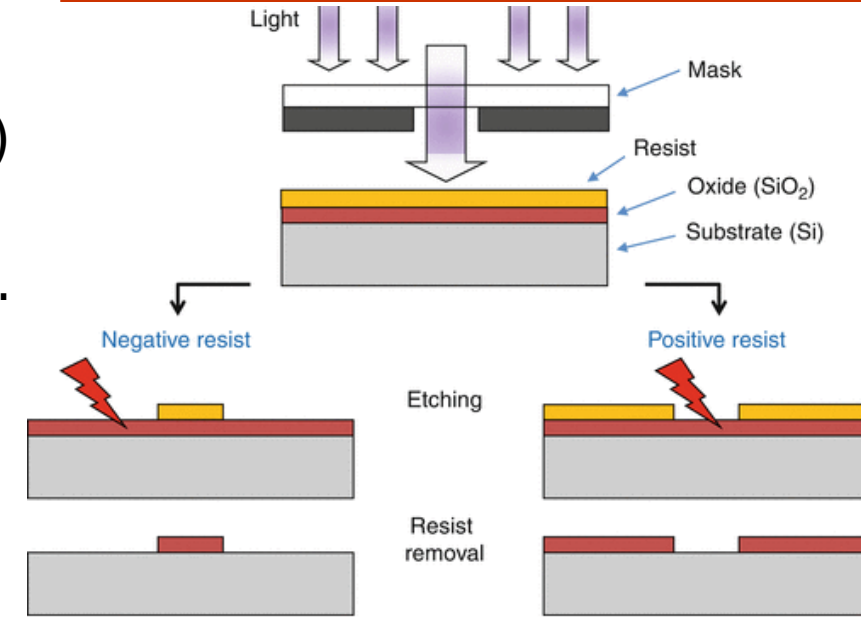
- ❑ 4004 → 4-bit CPU introduced in 1971
- ❑ # of transistors: 2300 with 10 μm technology.
- ❑ Clock: 740 kHz, Cache: 640 bytes
- ❑ Used in Scientific Calculator

- ❑ i9-12900KS → world's fastest desktop processor
- ❑ 64-bit Core i9 12th Gen, introduced in 2022
- ❑ # of transistors: billions with 7 nm technology.
- ❑ Cores: 16, Max Clock: 5.5 GHz, Cache: 30 Mbytes
- ❑ Used in High-performance computers.

Lithography

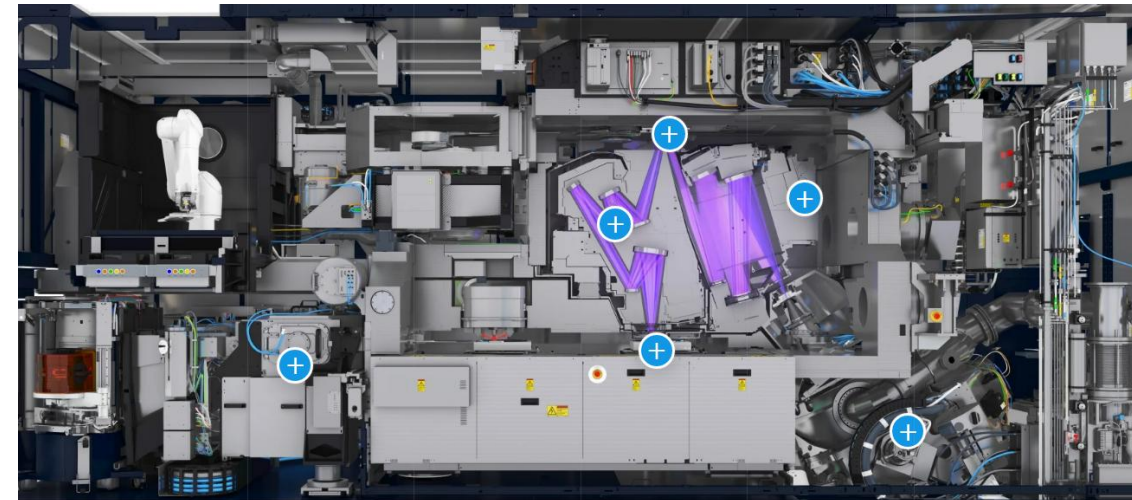
❑ ASML lithography System

- ❑ Deep Ultraviolet (**DUV**) lithography (> 7 nm process), Extreme Ultraviolet (**EUV**) lithography (**5 nm and 3 nm**)
- ❑ **TWINSCAN NXE:3600D** → The most advanced lithography system in the world. (5nm and 3nm)
- ❑ Currently, **ASML** is the only company that provides a commercial EUV system.
- ❑ Soon ASML will announce their 1 nm EUV lithography System.



❑ ASML Competitors

- ❑ Nikon, Canon, and Kioxia from Japan.
- ❑ SMEE from China.



<https://www.asml.com/en/products/euv-lithography-systems>

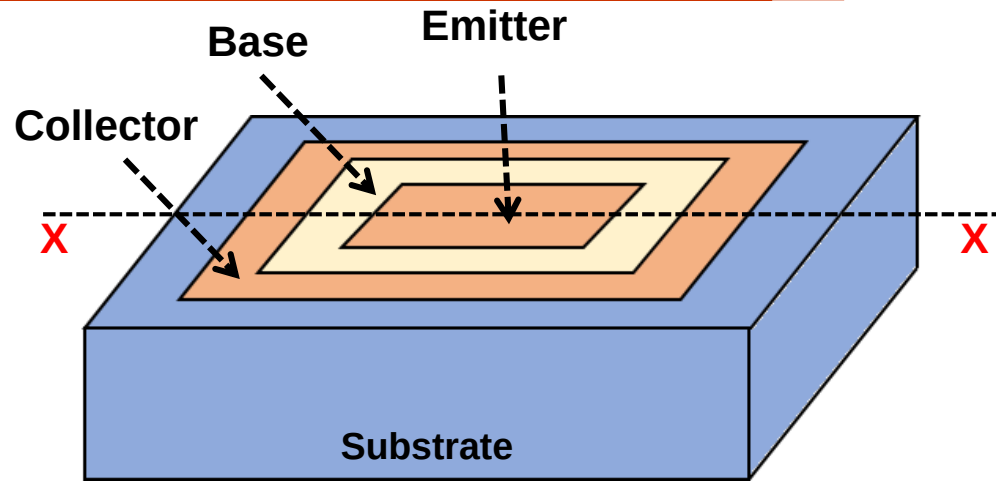
Check These Videos

- ❑ **Evolution of Transistor** → <https://www.youtube.com/watch?v=Z7M8etXUEUU>
- ❑ **Samsung Nanosheet transistor (MBCFET)** → <https://www.youtube.com/watch?v=3otqUu-7WUQ>
- ❑ **Intel Nanosheet transistor (RibbonFET)** → <https://www.youtube.com/watch?v=pymtOpvdRrl>
- ❑ **ASML lithography machines** → <https://www.youtube.com/watch?v=jL3TYXr52BY&list=PLLROOWd6snSJjAHlf9YvdjSF7DGxMDJsD&index=4>
- ❑ **A visit to ASML's cleanroom for EUV** → <https://www.youtube.com/watch?v=ttbaal5xUcg&list=PLLROOWd6snSJjAHlf9YvdjSF7DGxMDJsD&index=6>

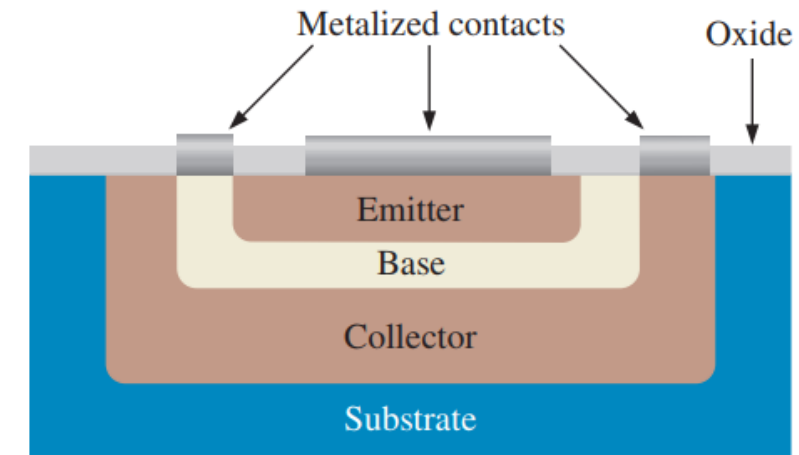
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BJT Construction



Cross-section XX



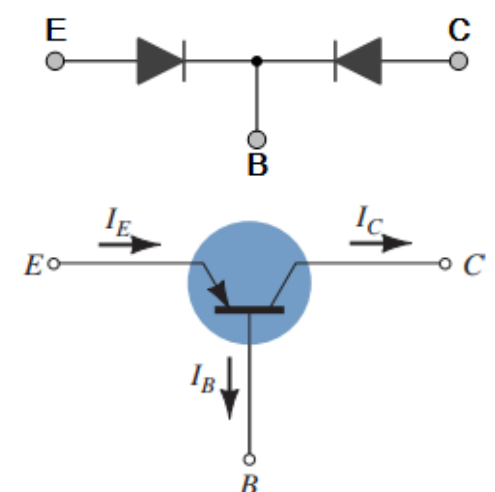
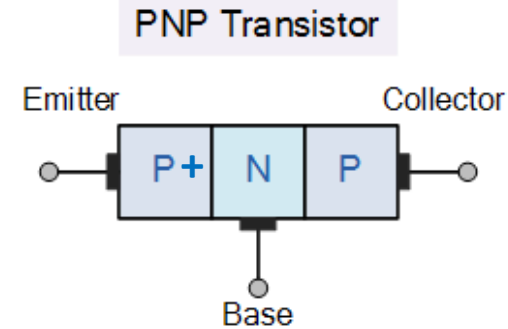
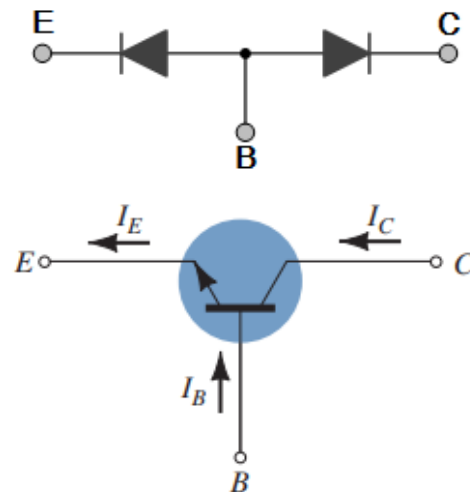
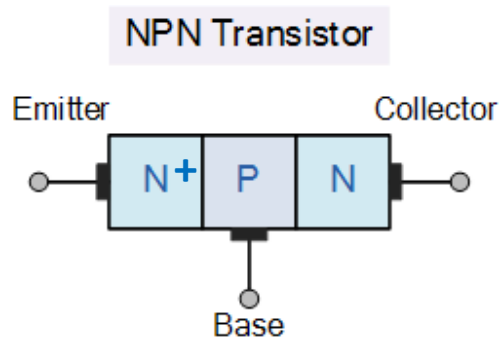
- ❑ The **BJT** is constructed with three doped semiconductor regions called **Emitter, Base, and Collector**.
- ❑ **Emitter** and **collector** have the **same** doping type, while the **base** has an **alternating** doping type → **npn** or **pnp** transistor
- ❑ **BJT current is due to the motion of both electrons and holes, therefore it is a Bi-polar device**

❑ Conditions for proper operation:

- ❑ Emitter should be heavily doped
- ❑ Base width should be as small as possible

❑ Why!!

BJT Operating Regions



Region	BEJ	BCJ	Application
Active	Forward	Reverse	Amplification
Saturation	Forward	Forward	Digital Switch (ON)
Cut off	Reverse	Reverse	Digital Switch (OFF)
Inverse Active	Reverse	Forward	In TTL gates.

BJT Operating Regions

Active:

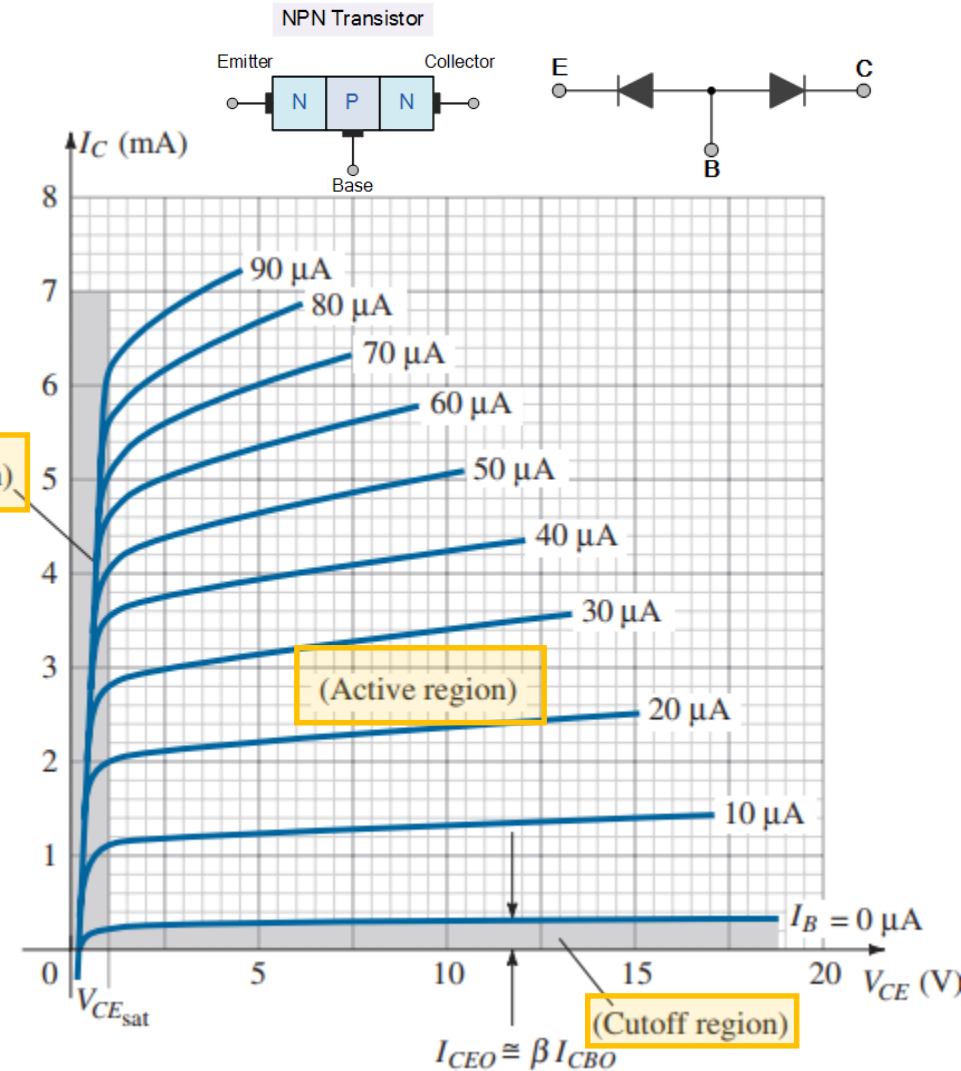
- conditions
- BEJ Forward $\rightarrow V_{BE} \approx 0.7 \text{ V}$
 - BCJ Reverse $\rightarrow V_{BC} < 0.5 \text{ V} \rightarrow V_{CE} > 0.2 \text{ V} (V_{CE_{sat}})$

- Valid equations
- $I_E = I_C + I_B$
 - $I_C = \beta I_B$
 - $I_C = \alpha I_E$
 - $I_C = I_s (e^{V_{BE}/V_T} - 1)$
 - $P \approx I_C V_{CE}$

- I_s : BJT reverse saturation current
- V_T : Thermal voltage (KT/q). About 26 mV at room temperature
- Relationship between Beta (β) and Alpha (α)

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$



Output Characteristics of npn transistor

BJT Operating Regions

❑ Saturation:

conditions

❑ BEJ Forward $\rightarrow V_{BE} \approx 0.7 \text{ V}$

❑ BCJ Forward $\rightarrow V_{BC} \approx 0.5 \text{ V} \rightarrow V_{CE} \approx 0.2 \text{ V} (V_{CE_{sat}})$

Valid equations

❑ $I_E = I_C + I_B$

❑ $P = I_C V_{CE} + I_B V_{BE}$

❑ Note:

❑ $I_C \neq \beta I_B$ ❑ $I_C < \beta I_B$

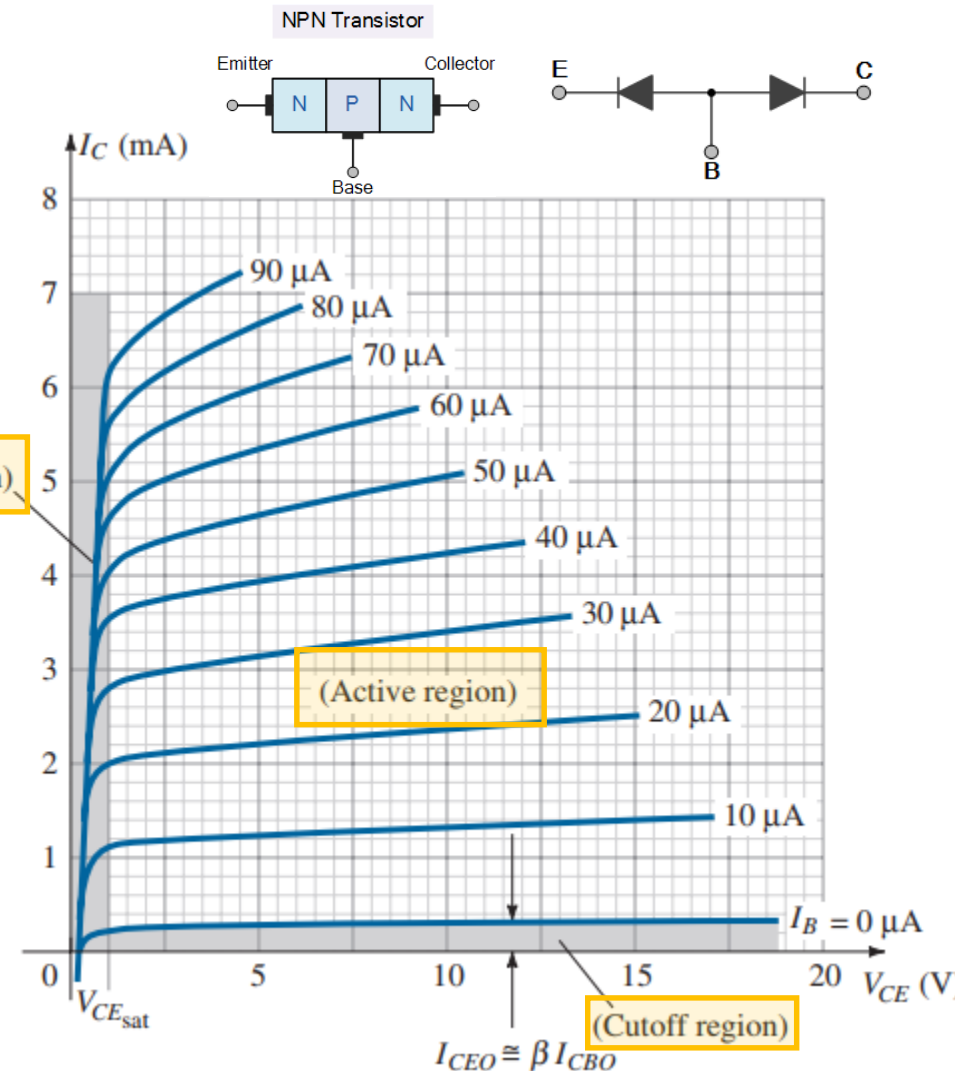
❑ In saturation, I_C is called $I_{C_{sat}}$: maximum possible collector current

❑ In saturation, $I_{C_{sat}} < \beta I_B \rightarrow I_B > \frac{I_{C_{sat}}}{\beta}$

(Saturation region)

(Active region)

(Cutoff region)



Output Characteristics of npn transistor

BJT Operating Regions

❑ **Cut off:**

conditions

❑ BEJ Reverse $\rightarrow V_{BE} < 0.7 \text{ V}$

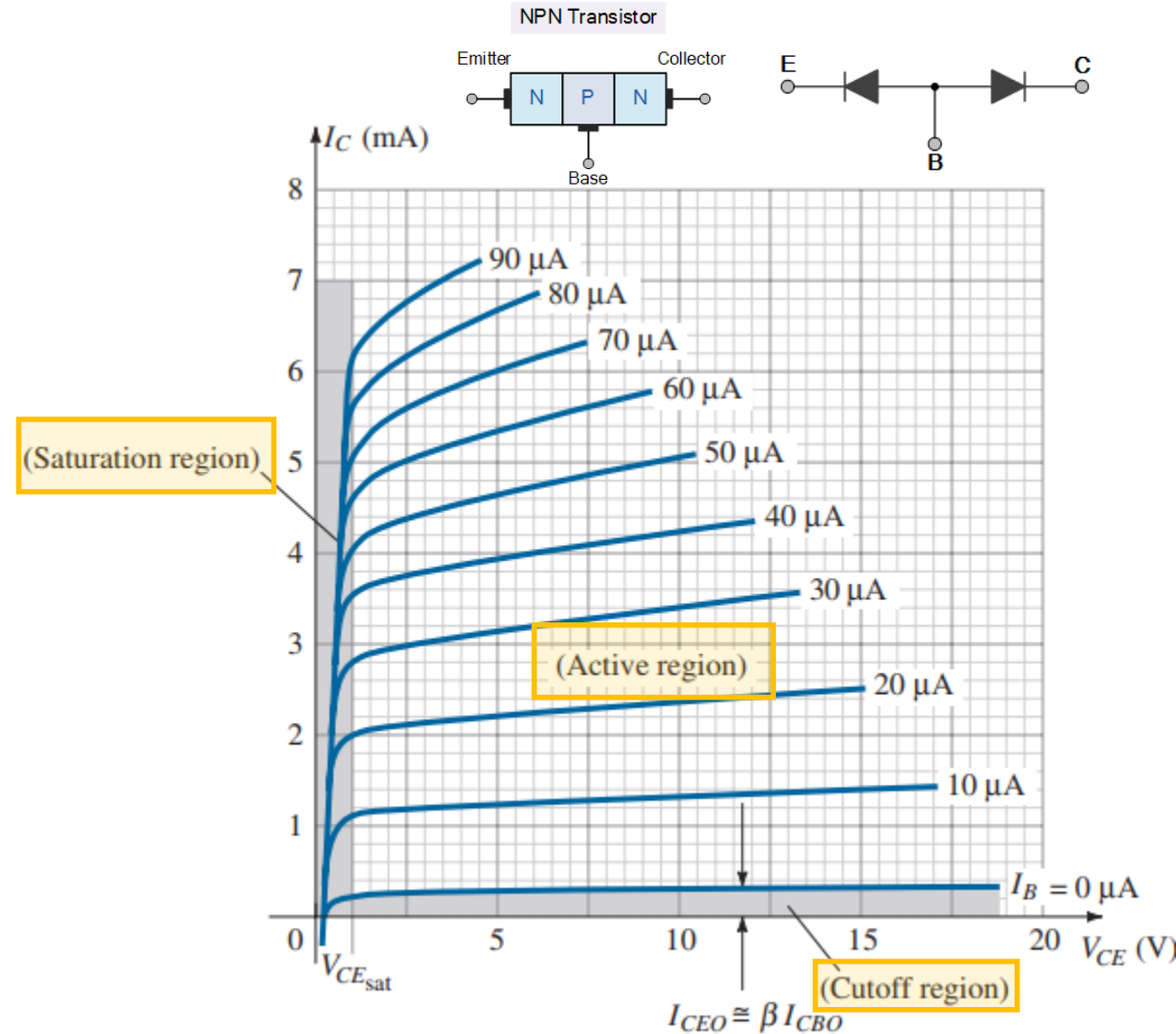
❑ BCJ Reverse $\rightarrow V_{BC} < 0.5 \text{ V}$

Valid
equations

❑ $I_B \approx 0$

❑ $I_C \approx I_E \approx I_{CE0} \approx 0$

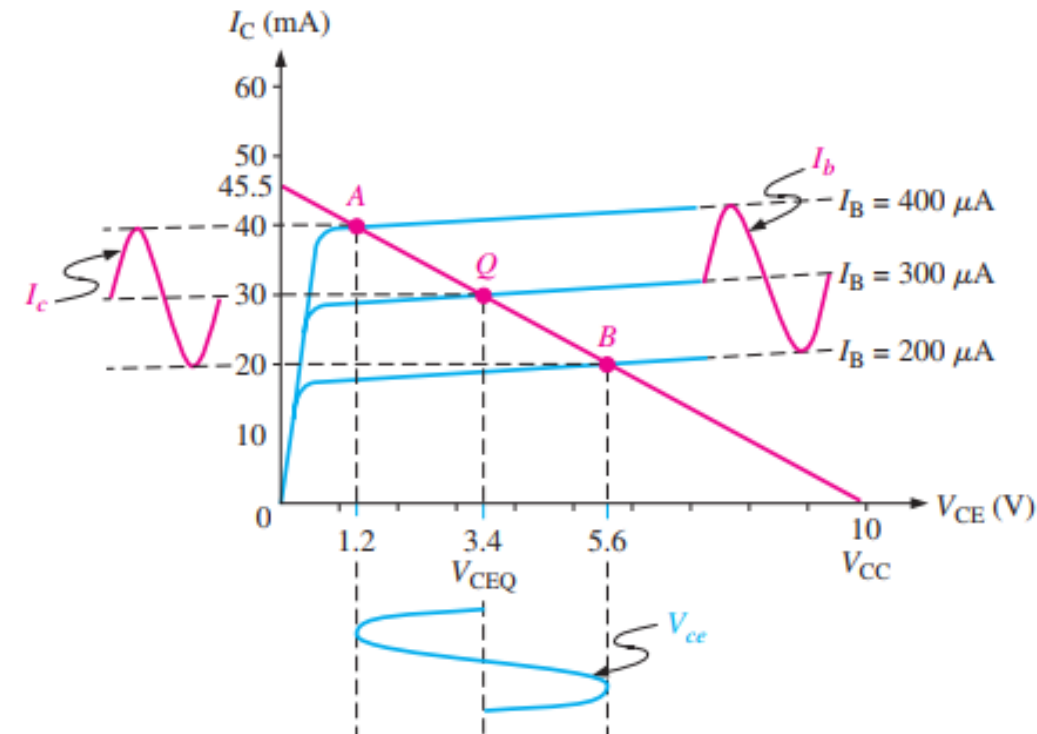
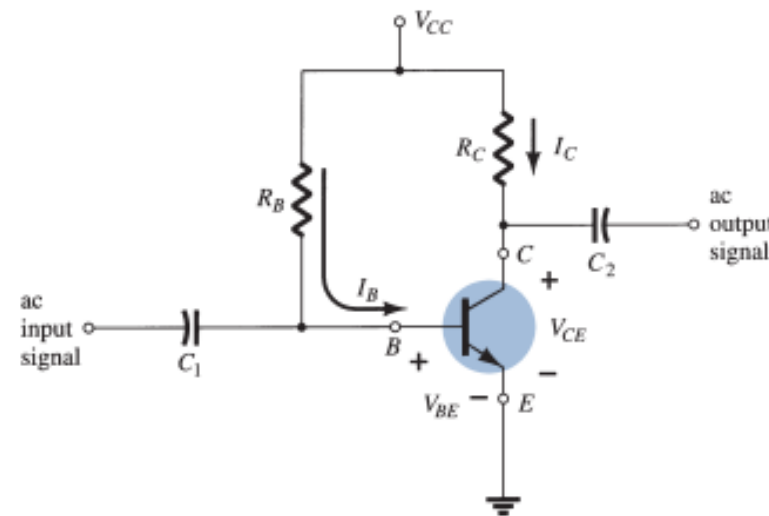
❑ $P \approx 0$



Output Characteristics of npn transistor

Transistor Amplification Action

- ❑ DC Supply and resistors are required to bias the transistor (Q-point) in the Active region such that the transistor acts as a linear amplifier. There are different biasing configurations that can be used to bias the transistor (Fixed-bias, Emitter-bias, Voltage divider, Collector feedback, etc.)
- ❑ What is the function of the coupling capacitor at the input and at the output?



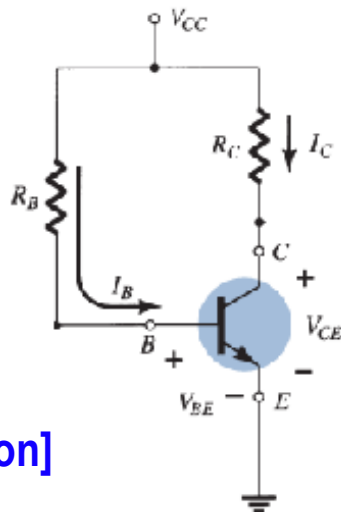
BJT DC Biasing Configurations

Fixed Bias

- Get I_B from Base-Emitter Loop

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \quad V_{BE} \approx 0.7 \text{ V}$$



- Get I_C [Assume operating in Active region]

$$I_C = \beta I_B = \beta \frac{V_{CC} - V_{BE}}{R_B}$$

- Get V_{CE} from Collector-Emitter Loop

$$V_{CC} - I_C R_C - V_{CE} = 0 \quad V_{CE} = V_{CC} - I_C R_C$$

Check

- If $V_{CE} > 0.2 \text{ V} \rightarrow \text{OK (in Active)}$
- If $V_{CE} < 0.2 \text{ V} \rightarrow \text{Not in active, transistor operate in saturation and } V_{CE} = 0.2 \text{ V. Then get } I_C = I_{C_{sat}} \text{ from collector-Emitter loop.}$

Emitter Bias

- Get I_B from Base-Emitter Loop

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\text{Sub. with } I_E = (\beta + 1)I_B$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \quad V_{BE} \approx 0.7 \text{ V}$$

- Get I_C [Assume operating in Active region]

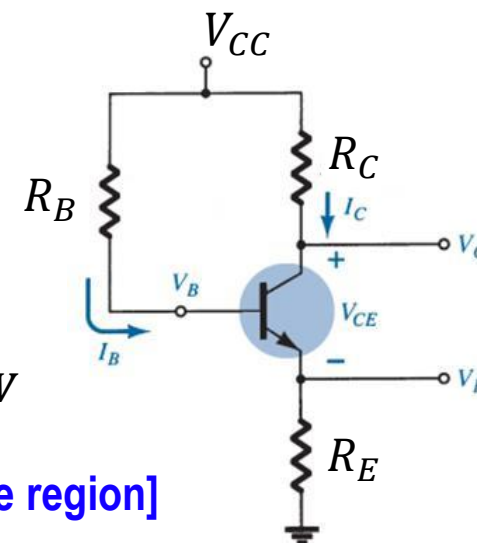
$$I_C = \beta I_B = \frac{\beta(V_{CC} - V_{BE})}{R_B + (\beta + 1)R_E} \approx \frac{V_{CC} - V_{BE}}{\frac{R_B}{\beta} + R_E}$$

- Get V_{CE} from Collector-Emitter Loop

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Check !!



Better stability in the Q-point as β changes with temperature

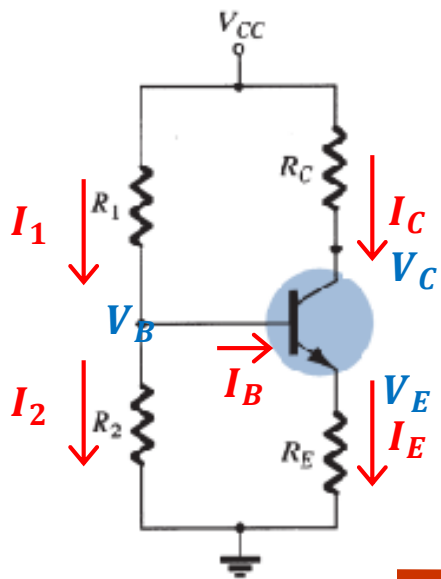
Sub. with

$$I_E = I_C / \alpha \approx I_C$$

BJT DC Biasing Configurations

❑ **Exact**

❑ **Voltage Divider**



R_{th}, E_{th}

I_B, I_C

V_{CE}

- ❑ Get equivalent Thevenin circuit seen at the Base, Then get I_B, I_C, V_{CE}

$$R_{th} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

$$E_{th} = V_{CC} \frac{R_2}{R_1 + R_2}$$

$$E_{th} - I_B R_{th} - V_{BE} - I_E R_E = 0$$

Sub. With $I_E = (\beta + 1)I_B$

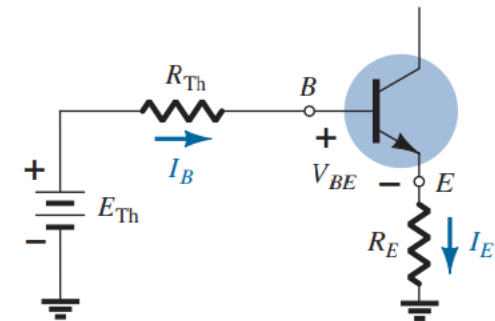
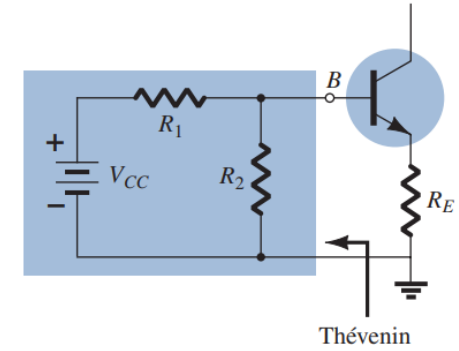
$$I_B = \frac{E_{th} - V_{BE}}{R_{th} + (\beta + 1)R_E}$$

$$I_C = \beta I_B = \frac{\beta (E_{th} - V_{BE})}{R_{th} + (\beta + 1)R_E} \approx \frac{E_{th} - V_{th}}{\frac{R_{th}}{\beta} + R_E}$$

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

Sub. with $I_E \approx I_C$

$$V_{CE} = V_{CC} - I_C (R_C + R_E) \quad \text{Check !!}$$



❑ **Approximate Analysis:** ($\beta R_E > 10R_2$)

Ignore I_B w.r.t I_2

$$V_B = V_{CC} \frac{R_2}{R_1 + R_2}$$

$$V_E = V_B - V_{BE}$$

$$I_E = \frac{V_E}{R_E}$$

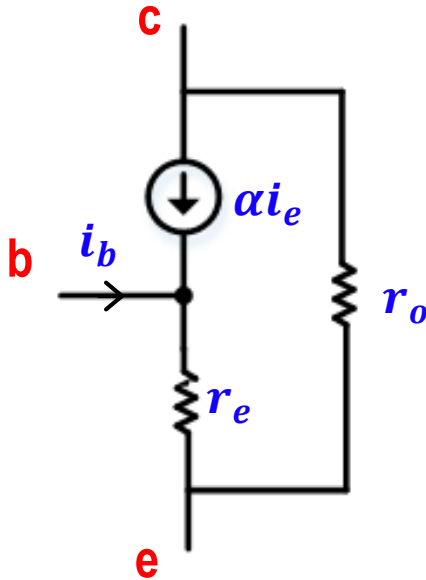
$$I_C \approx I_E$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

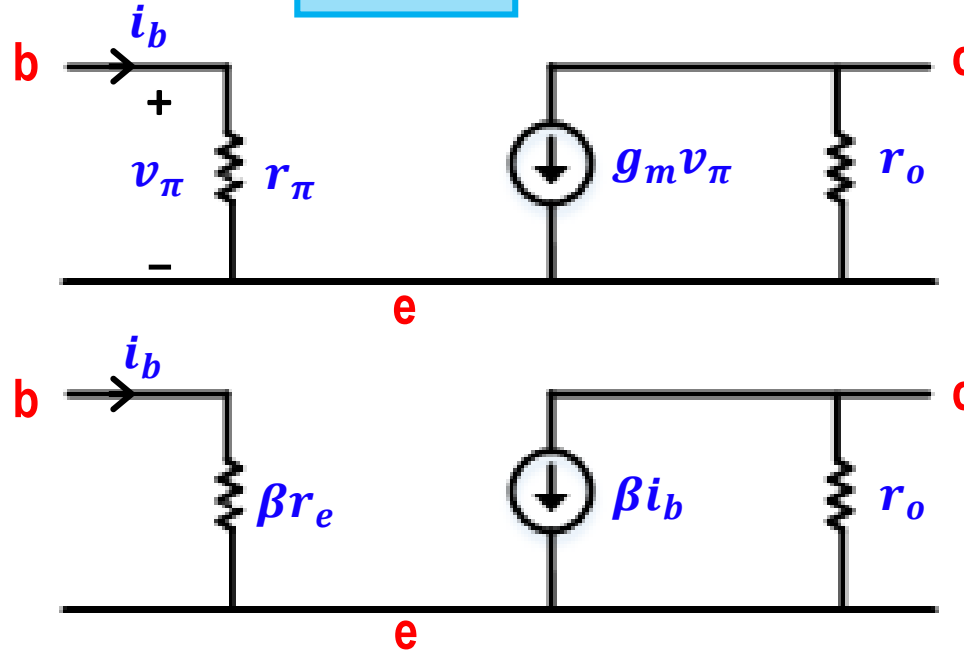
❑ Voltage divider bias provides a better Q-point stability compared to Emitter bias

BJT Small Signal Model

T Model



π Model



$$r_e = \frac{V_T}{I_E} = \frac{26\text{mV}}{I_E}$$

$$r_\pi = \beta r_e = \beta \frac{V_T}{I_E} = \frac{V_T}{I_B}$$

g_m : transistor transconductance

$$I_C \approx I_S e^{V_{BE}/V_T}$$

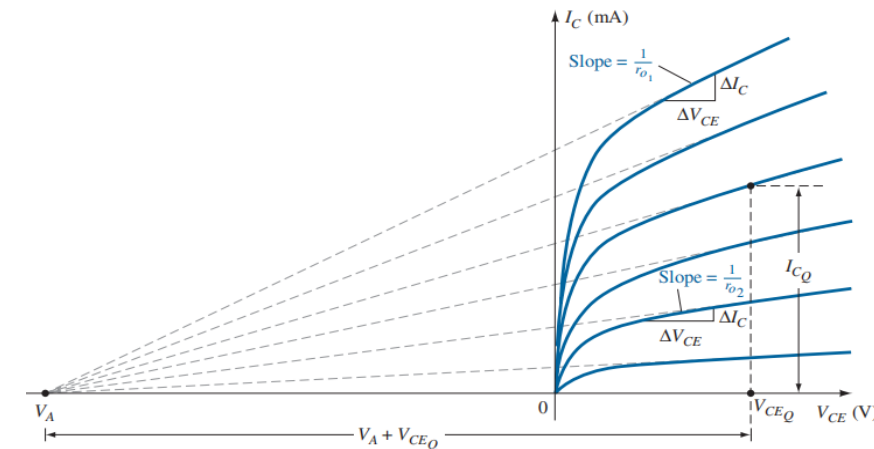
$$g_m = \frac{\partial I_C}{\partial V_{BE}} = I_S \frac{e^{V_{BE}/V_T}}{V_T} = \frac{I_C}{V_T}$$

$$g_m \approx \frac{1}{r_e}$$

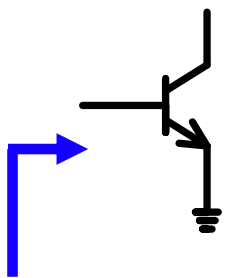
r_o : Output impedance

$$r_o = \frac{\partial V_{CE}}{\partial I_C} = \frac{V_A + V_{CEQ}}{I_{CQ}} \approx \frac{V_A}{I_{CQ}}$$

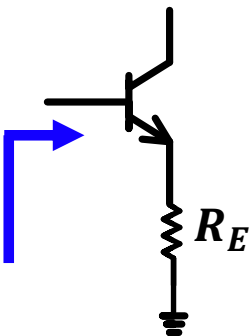
V_A is the early voltage



Impedance Shortcuts

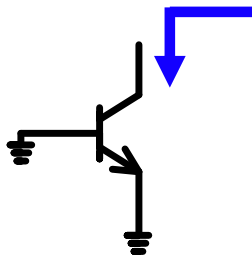


$$Z = r_{\pi}$$

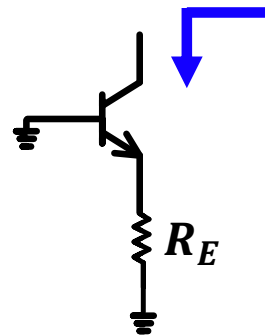


$$Z = r_{\pi} + \beta R_E$$

$$= r_{\pi} [1 + g_m R_E]$$

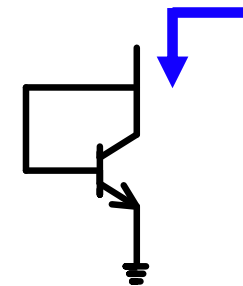


$$Z = r_o$$



$$Z = r_o + (1 + g_m r_o)(R_E \parallel r_{\pi})$$

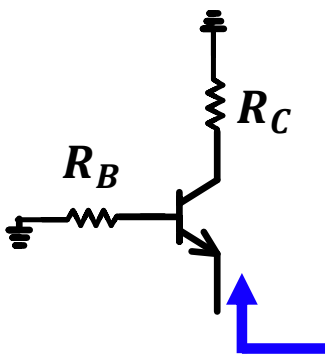
$$\approx r_o [1 + g_m (R_E \parallel r_{\pi})]$$



$$Z = r_{\pi} \parallel \frac{1}{g_m} \parallel r_o$$

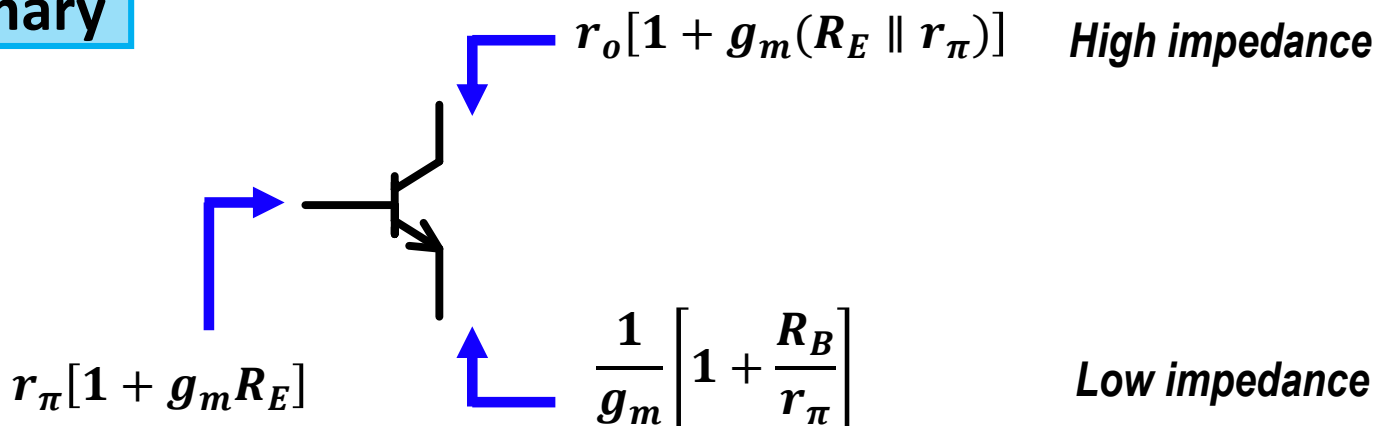
$$\approx \frac{1}{g_m}$$

Summary



$$Z = \frac{r_{\pi} + R_B}{1 + g_m r_{\pi}}$$

$$\approx \frac{1}{g_m} \left[1 + \frac{R_B}{r_{\pi}} \right]$$

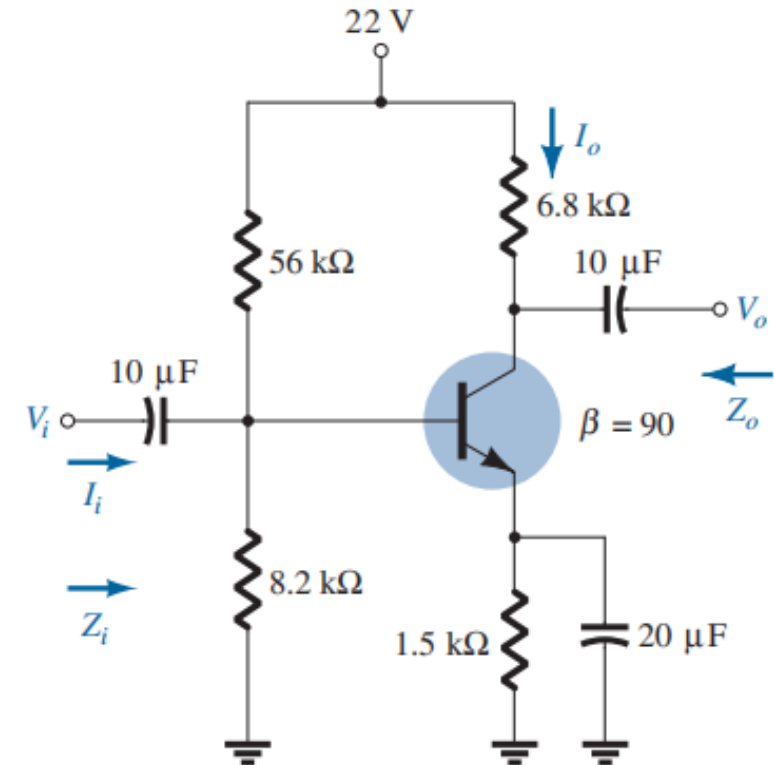


High impedance

Low impedance

Example # 1 (1/7)

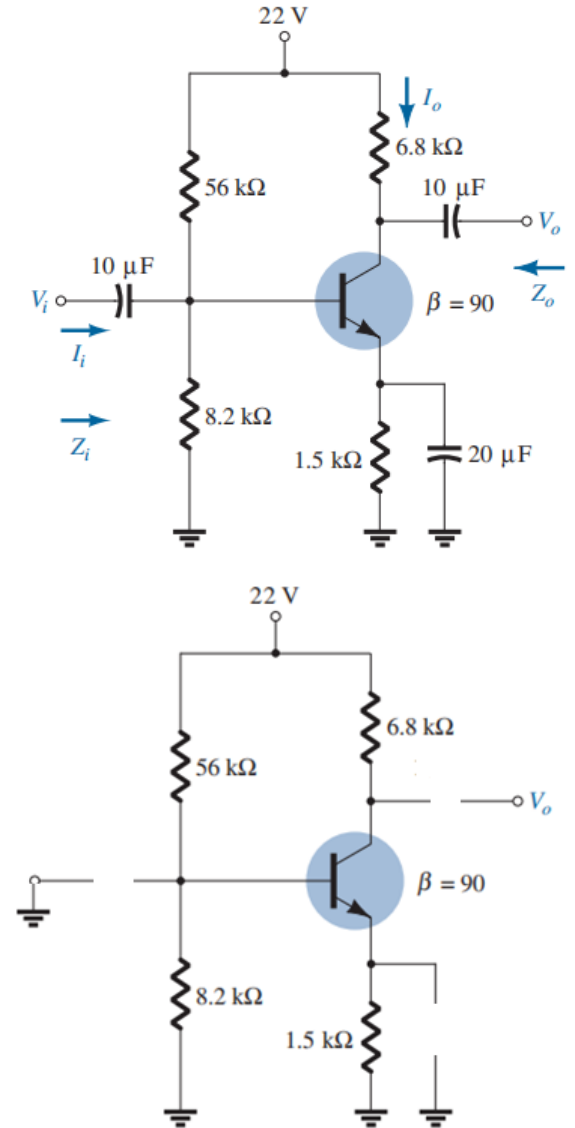
- ☐ For the shown BJT amplifier circuit, find the following. Note ($\beta = 90$ and $V_A = 80\text{ V}$)
 - ☐ Q-point
 - ☐ Z_i
 - ☐ Z_o
 - ☐ A_v
 - ☐ Sketch the input and output waveforms if the input is a sinusoidal signal with 20 mVpp (peak-to-peak)



Example #1 (2/7)

DC Analysis

- ❑ Cancel all AC sources
 - ❑ AC VS (S.C), AC CS (O.C)
- ❑ At DC $\rightarrow f = 0 \text{ Hz}$
 - ❑ Capacitors: $X_C = 1 / 2\pi f C = 1 / 0 = \infty \Omega$ (O.C)
 - ❑ Inductors: $X_L = 2\pi f L = 0 \Omega$ (S.C)
- ❑ Solve input and output loops to get the Q-point



Example #1 (3/7)

□ Check for approximate

$$\square \beta R_E > 10R_2 \rightarrow 90(1.5K\Omega) > 10(8.2K\Omega) \rightarrow 135K\Omega > 82K\Omega \text{ (satisfied)}$$

$$V_B = V_{CC} \frac{R_2}{R_1 + R_2} = 2.81 \text{ V} \quad V_E = V_B - V_{BE} = 2.81 \text{ V} - 0.7 \text{ V} = 2.11 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{2.11 \text{ V}}{1.5K\Omega} = 1.41 \text{ mA} \quad I_C \approx I_E = 1.41 \text{ mA}$$

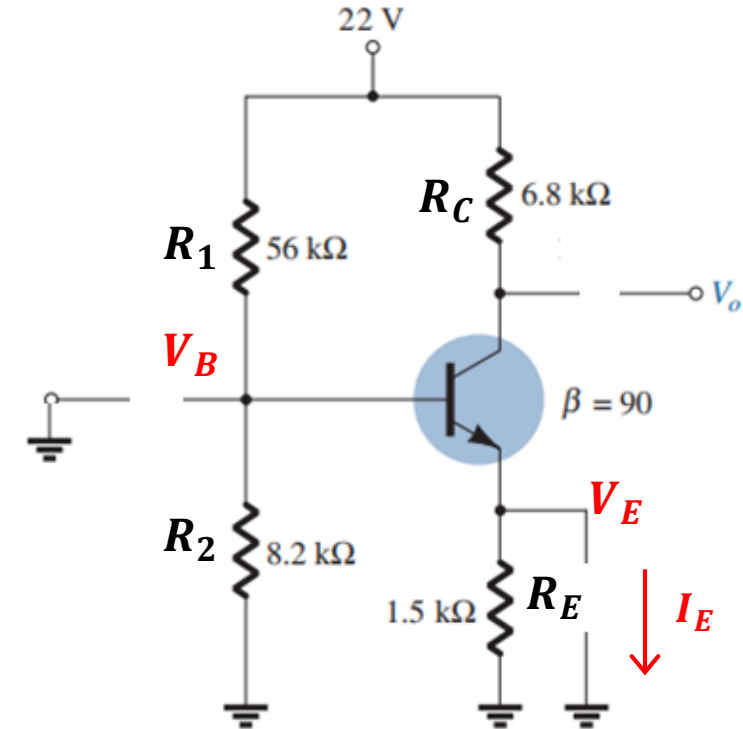
$$\text{Output loop} \rightarrow V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) = 22 \text{ V} - 1.41 \text{ mA}(6.8K\Omega + 1.5K\Omega) = 10.297 \text{ V}$$

$$V_C = V_{CC} - I_C R_C = 22 \text{ V} - 1.41 \text{ mA}(6.8K\Omega) = 12.41 \text{ V}$$

Check to be in active $\rightarrow$ OK as $V_{CE} > 0.2 \text{ V}$

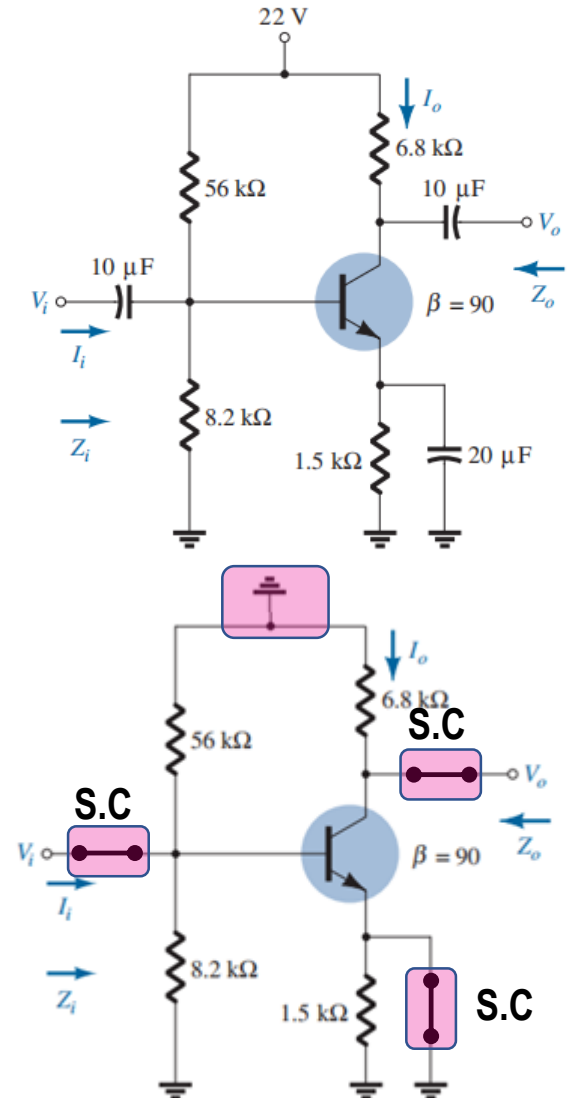
Q-point: $I_C = 1.41 \text{ mA}$, $V_{CE} = 10.297 \text{ V}$



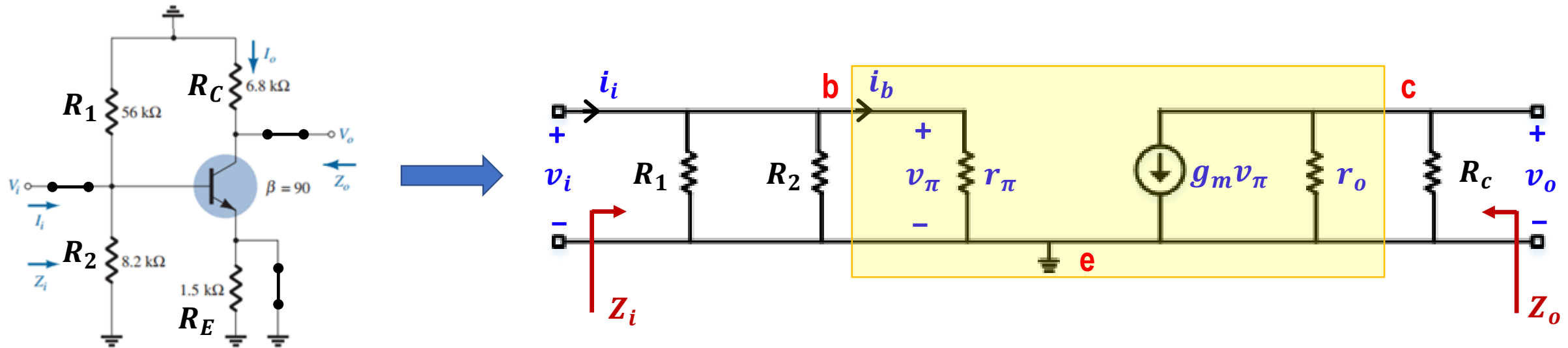
Example #1 (4/7)

AC Analysis

- ❑ Cancel all DC sources
 - ❑ DC VS (S.C), DC CS (O.C)
- ❑ At Mid and high band frequencies $\rightarrow f \uparrow\uparrow$
 - ❑ Coupling Capacitors: $X_C = 1 / 2\pi f C \approx 0 \Omega$ (S.C)
- ❑ Draw the small signal equivalent circuit
- ❑ Get r_e, r_π, g_m, r_o
- ❑ Calculate $Z_i, Z_o, A_V, \dots$



Example #1 (5/7)



□ Get r_e , r_π , g_m , r_o

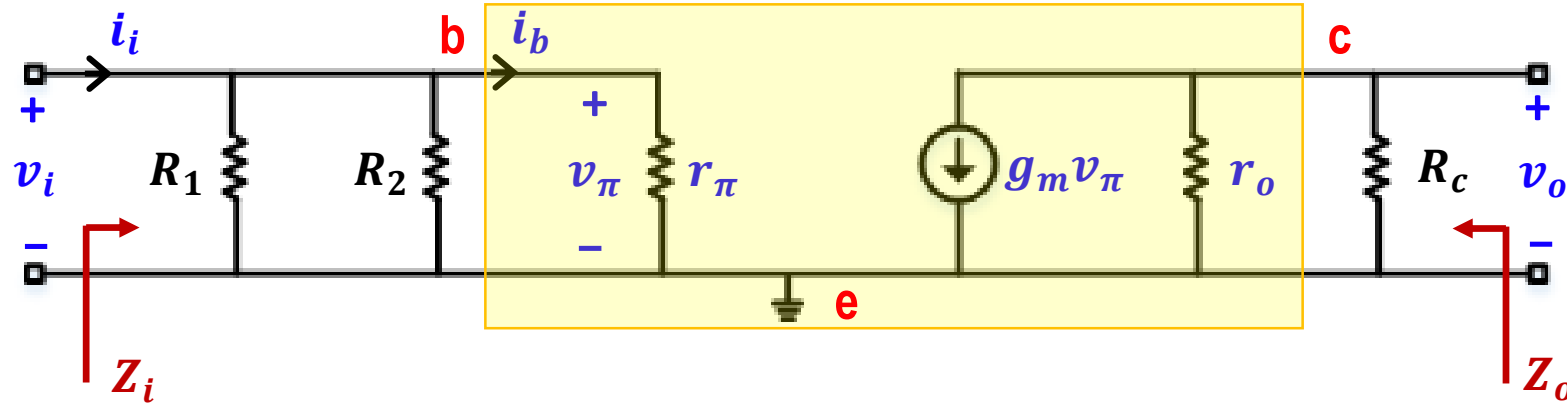
$$\triangleright r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{1.41\text{ mA}} = 18.44\ \Omega$$

$$\triangleright r_\pi = \beta r_e = 90 \times 18.44 = 1.66\text{ K}\Omega$$

$$\triangleright g_m = \frac{I_C}{V_T} = 54.23\text{ mS}$$

$$\triangleright r_o = \frac{V_A}{I_C} = \frac{80\text{ V}}{1.41\text{ mA}} = 56.74\text{ K}\Omega$$

Example #1 (6/7)



$$\square Z_i = \frac{v_i}{i_i} = R_1 \parallel R_2 \parallel r_\pi = 56 \text{ K}\Omega \parallel 8.2 \text{ K}\Omega \parallel 1.66 \text{ K}\Omega = 1.347 \text{ K}\Omega$$

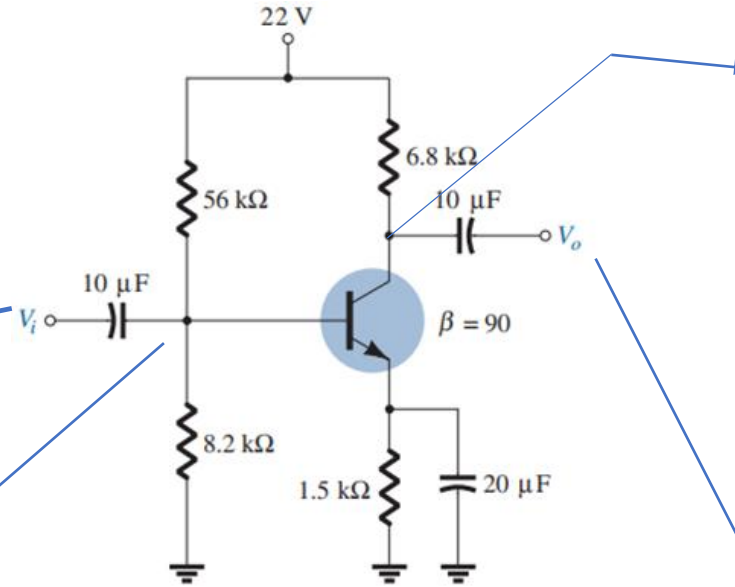
$$\square Z_o = \left. \frac{v_o}{i_o} \right|_{v_i=0} \quad \text{At } v_i = 0 \rightarrow v_\pi = 0 \rightarrow g_m v_\pi = 0 \text{ (open circuit current source)}$$

$$\blacktriangleright Z_o = r_o \parallel R_C = 56.74 \text{ K}\Omega \parallel 6.8 \text{ K}\Omega = 6.07 \text{ K}\Omega$$

$$\square A_v = \frac{v_o}{v_i} = \frac{-g_m V_\pi (r_o \parallel R_C)}{V_\pi} = -g_m (r_o \parallel R_C) = -(54.23 \text{ mS}) \times (6.07 \text{ K}\Omega) = -329.18 \text{ V/V}$$

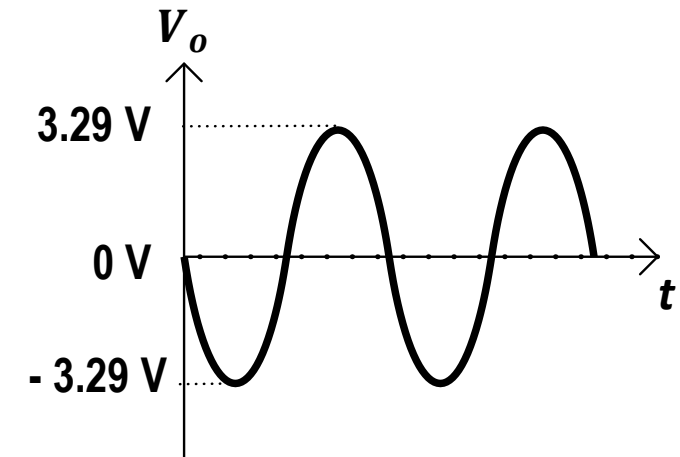
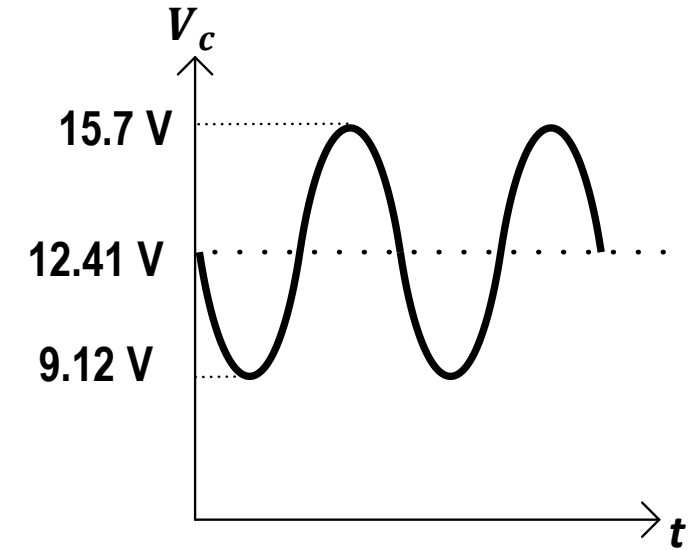
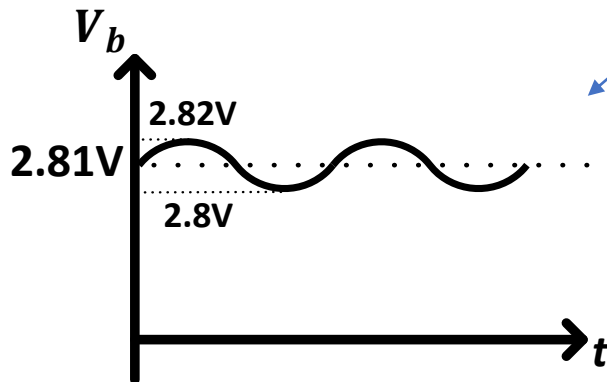
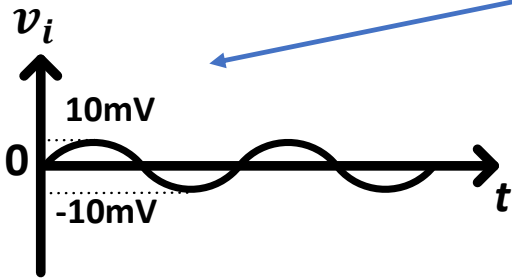
Example #1 (7/7)

- Sketch the input and output waveforms if the input is a sinusoidal signal with 20 mVpp



- $v_o = |A_v| \times v_i = 329.18 \times 20 \text{ mVpp} = 6.58 \text{ Vpp}$

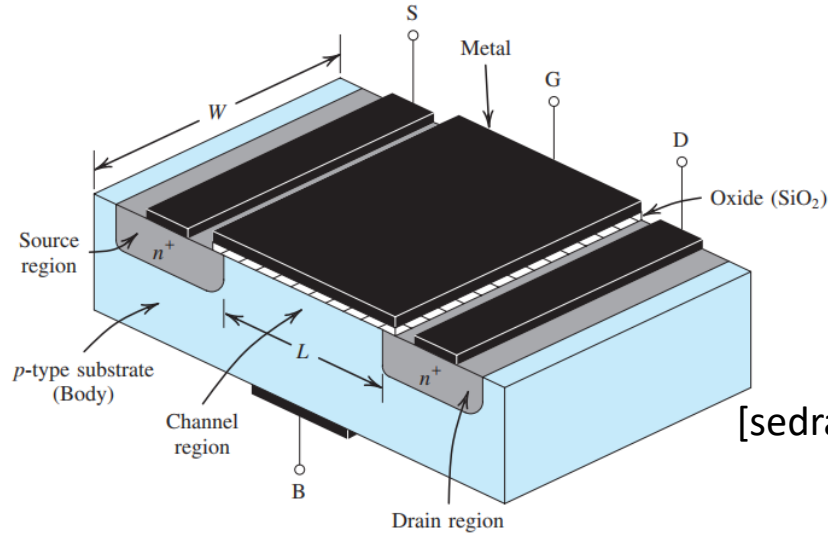
- (Q) repeat if the input signal amplitude is 100 mVpp



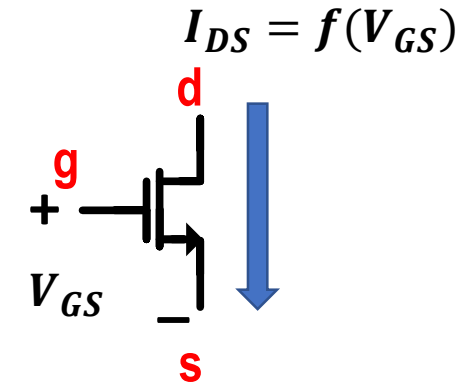
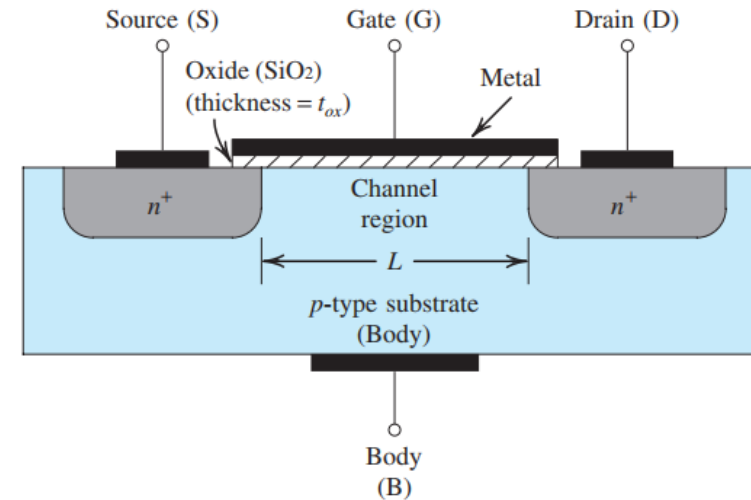
Outline

- ☐ Course Information
- ☐ Introduction: Semiconductors between past, present, and future
- ☐ Quick Review
 - ☐ Bipolar Junction Transistor (BJT)
 - ☐ Metal Oxide Semiconductor Field Effect Transistor (MOSFET)
- ☐ Amplifier Model

MOSFET Construction



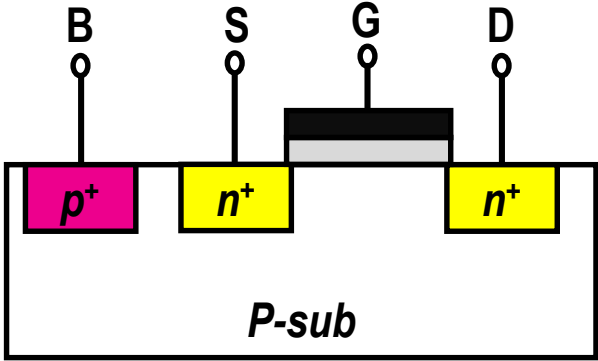
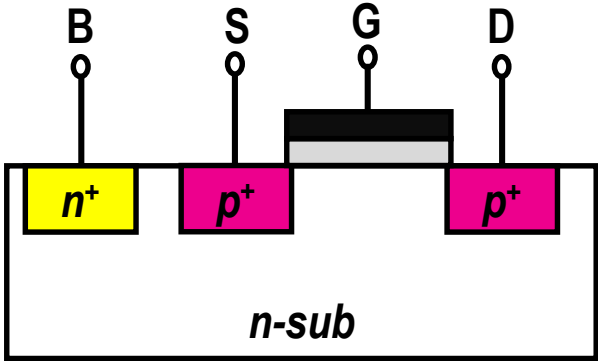
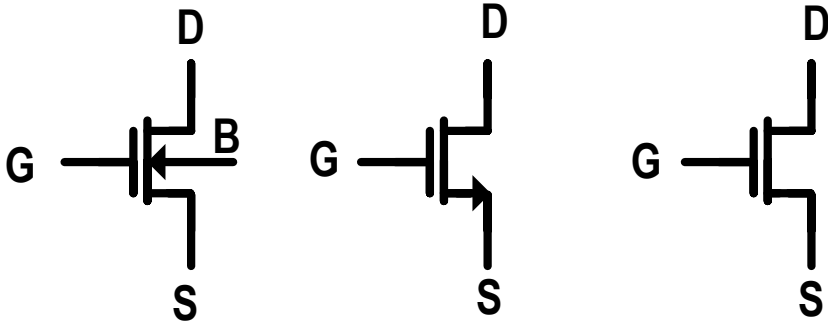
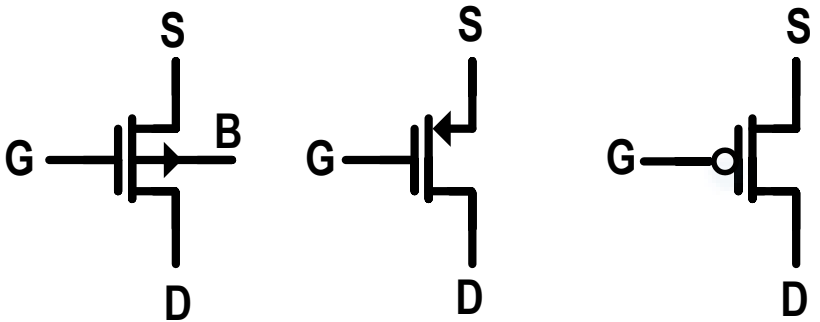
[sedra, 2014]



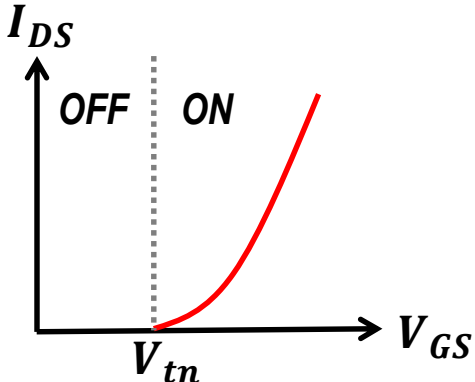
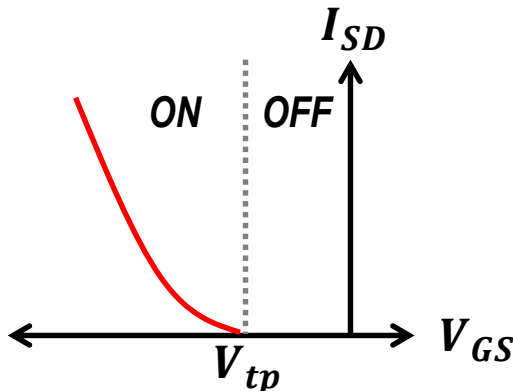
- ❑ **MOSFET** is a **unipolar** device → device current is due to either electrons (**NMOS**) or holes (**PMOS**)
- ❑ **MOSFET** acts as a **VCCS** → the current between the Drain and source depends on the voltage between gate and source
- ❑ **Gate**: formed of metal (45 nm process and onward) or poly silicon
- ❑ **Dielectric**: Silicon oxide (SiO₂) or high-dielectric (high-k) materials. Thickness ~ 1 nm – 10 nm
- ❑ **Channel length**: $L \rightarrow 10\ \mu\text{m} - 3\ \text{nm}$, and much shorter in the future
- ❑ **Channel width** : $W \rightarrow 100\ \mu\text{m} - 50\ \text{nm}$

❑ **Note**: $I_G = 0$

NMOS and PMOS

	NMOS	PMOS
Construction		
Symbols		

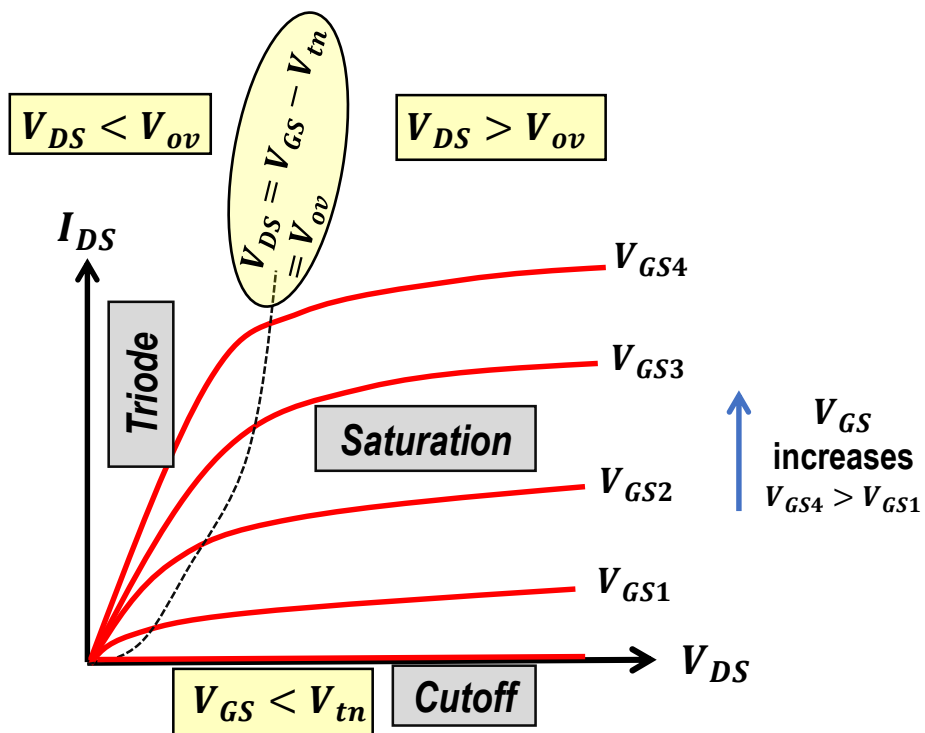
NMOS and PMOS

	NMOS	PMOS
Transfer Curve	 ☐ V_{tn}: Threshold voltage of NMOS ☐ V_{tn} is + ve for enhancement type and – ve for depletion type. ☐ If $V_{GS} > V_{tn}$, NMOS is ON and current flows from D to S ☐ If $V_{GS} < V_{tn}$, NMOS is OFF and $I_{DS} = 0$	 ☐ V_{tp}: Threshold voltage of PMOS ☐ V_{tp} is – ve for enhancement type and + ve for depletion type. ☐ If $V_{GS} < V_{tp}$, PMOS is ON and a current flows from S to D ☐ If $V_{GS} > V_{tp}$, PMOS is OFF and $I_{SD} = 0$

NMOS and PMOS

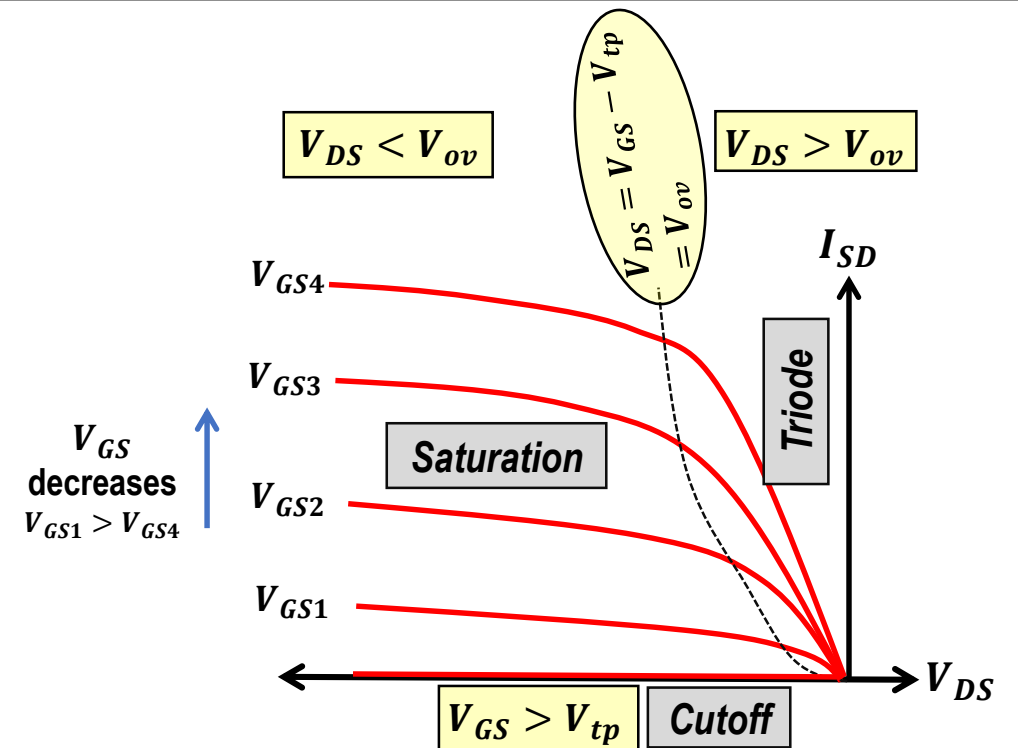
Output ch's Curve

NMOS



- ❑ Overdrive voltage (V_{ov}) is the difference between V_{GS} and V_t ($V_{ov} = V_{GS} - V_{tn}$)
- ❑ For NMOS to be ON $\rightarrow V_{GS} > V_{tn} \rightarrow V_{ov} > 0$

PMOS

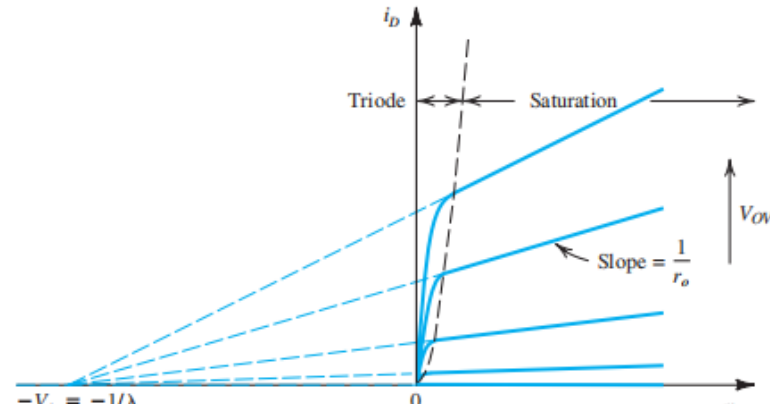


- ❑ For PMOS to be ON $\rightarrow V_{GS} < V_{tp} \rightarrow V_{ov} < 0$
- ❑ Note: V_{GS}, V_{DS}, V_{ov} of PMOS are negative during ON state.

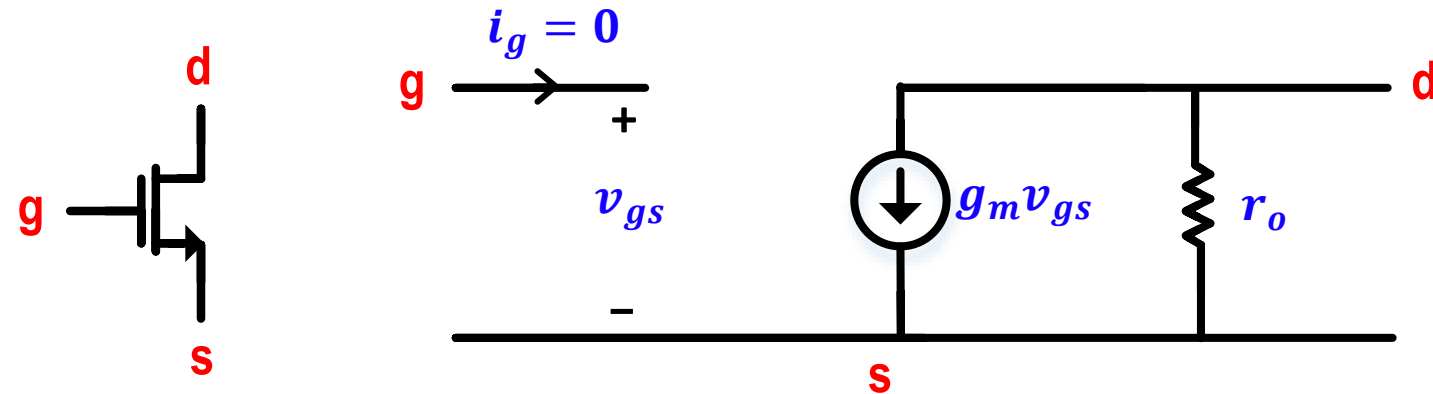
NMOS and PMOS

	NMOS	PMOS
Current Equation	Cutoff : $V_{GS} < V_{tn}$ □ $I_{DS} = 0$ Triode : $V_{GS} > V_{tn}$ and $V_{DS} < V_{ov}$ □ $I_{DS} = \beta_n \left[V_{ov} V_{DS} - \frac{V_{DS}^2}{2} \right]$ Saturation : $V_{GS} > V_{tn}$ and $V_{DS} > V_{ov}$ □ $I_{DS} = \frac{\beta_n}{2} (V_{ov})^2 \times (1 + \lambda_n V_{DS})$	Cutoff : $V_{GS} > V_{tp} \rightarrow V_{GS} < V_{tp}$ □ $I_{SD} = 0$ Triode : $V_{GS} < V_{tp}$ and $V_{DS} > V_{ov}$ Equivalent to Triode : $V_{GS} > V_{tp}$ and $V_{DS} < V_{ov}$ □ $I_{SD} = \beta_p \left[V_{ov} V_{DS} - \frac{V_{DS}^2}{2} \right] = \beta_p \left[V_{ov} V_{DS} - \frac{V_{DS}^2}{2} \right]$ Saturation : $V_{GS} < V_{tp}$ and $V_{DS} < V_{ov}$ Equivalent to Saturation : $V_{GS} > V_{tp}$ and $V_{DS} > V_{ov}$ □ $I_{SD} = \frac{\beta_p}{2} (V_{ov})^2 \times (1 + \lambda_p V_{DS}) = \frac{\beta_p}{2} (V_{ov})^2 \times (1 + \lambda_p V_{DS})$

NMOS and PMOS

	NMOS	PMOS
Beta	$B_n = \mu_n C_{ox} \frac{W}{L}$ $[A/V^2]$ μ_n : Electrons mobility $[m^2/V.s]$ C_{ox} : Oxide capacitance per unit area $[F/m^2]$ $C_{ox} = \frac{\epsilon_o \epsilon_r}{t_{ox}}$ ϵ_o: Air permittivity (8.854×10^{-12} F/m) ϵ_r: Sio2 relative permittivity (3.9)	$B_p = \mu_p C_{ox} \frac{W}{L}$ $[A/V^2]$ μ_p : holes mobility $[m^2/V.s]$ - As $\mu_n > \mu_p \rightarrow \beta_n > \beta_p$
Lambda	λ : Channel length modulation coefficient $[V^{-1}]$ $\lambda_n [+Ve], \lambda_p [-Ve]$ $\lambda = \frac{1}{V_A}$ V_A : Early voltage $r_o = \frac{V_A}{I_D} = \frac{1}{\lambda I_D}$ - Note: At $\lambda = 0 \rightarrow V_A = \infty \rightarrow r_o = \infty$	

MOSFET Small Signal Model



g_m : transistor transconductance

$$I_{DS} = \frac{\beta_n}{2} (V_{ov})^2 = \frac{\beta_n}{2} (V_{gs} - V_{tn})^2$$

$$g_m = \frac{\partial I_{DS}}{\partial V_{GS}} = \beta_n (V_{GS} - V_{tn}) = \beta_n V_{ov}$$

r_o : Output impedance

$$r_o = \frac{\partial V_{DS}}{\partial I_{DS}} \approx \frac{V_A}{I_{DS}} = \frac{1}{\lambda I_{DS}}$$

$$g_m = \beta_n V_{ov}$$

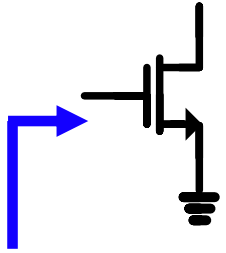
Sub. for $V_{ov} = \sqrt{\frac{2 I_{DS}}{\beta_n}}$

$$g_m = \sqrt{2 \beta_n I_{DS}}$$

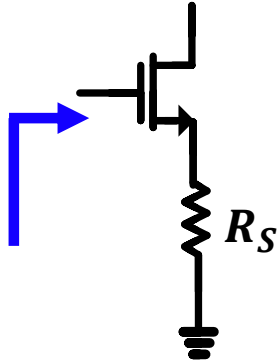
$$g_m = \frac{\beta_n V_{ov}^2}{V_{ov}}$$

$$g_m = \frac{2 I_D}{V_{ov}}$$

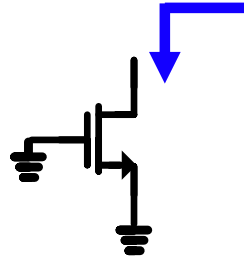
Impedance Shortcuts



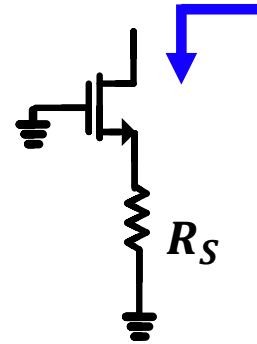
$$Z = \infty$$



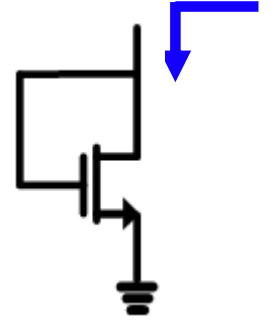
$$Z = \infty$$



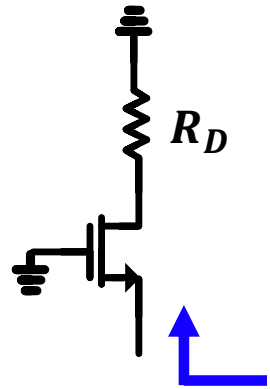
$$Z = r_o$$



$$Z = r_o + (1 + g_m r_o)(R_S) \\ \approx r_o[1 + g_m R_S]$$



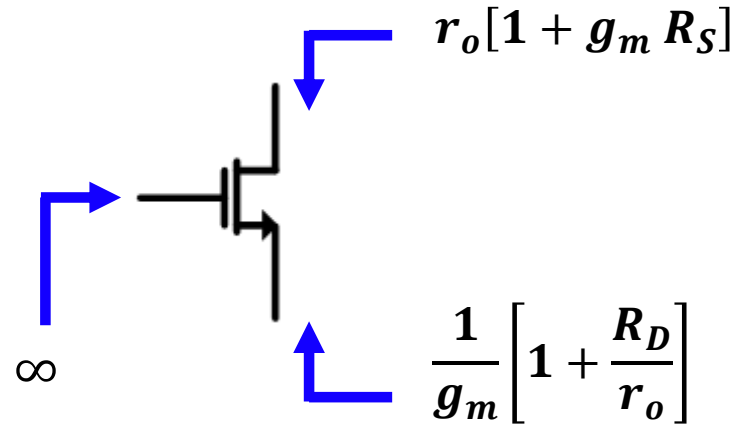
$$Z = \frac{1}{g_m} \parallel r_o \approx \frac{1}{g_m}$$



$$Z = \frac{R_D + r_o}{1 + g_m r_o} \\ \approx \frac{1}{g_m} \left[1 + \frac{R_D}{r_o} \right]$$

$$\text{If } R_D = 0, Z = \frac{1}{g_m}$$

Summary



$$r_o[1 + g_m R_S]$$

High impedance

$$\frac{1}{g_m} \left[1 + \frac{R_D}{r_o} \right]$$

Low impedance

Example #2 (1/7)

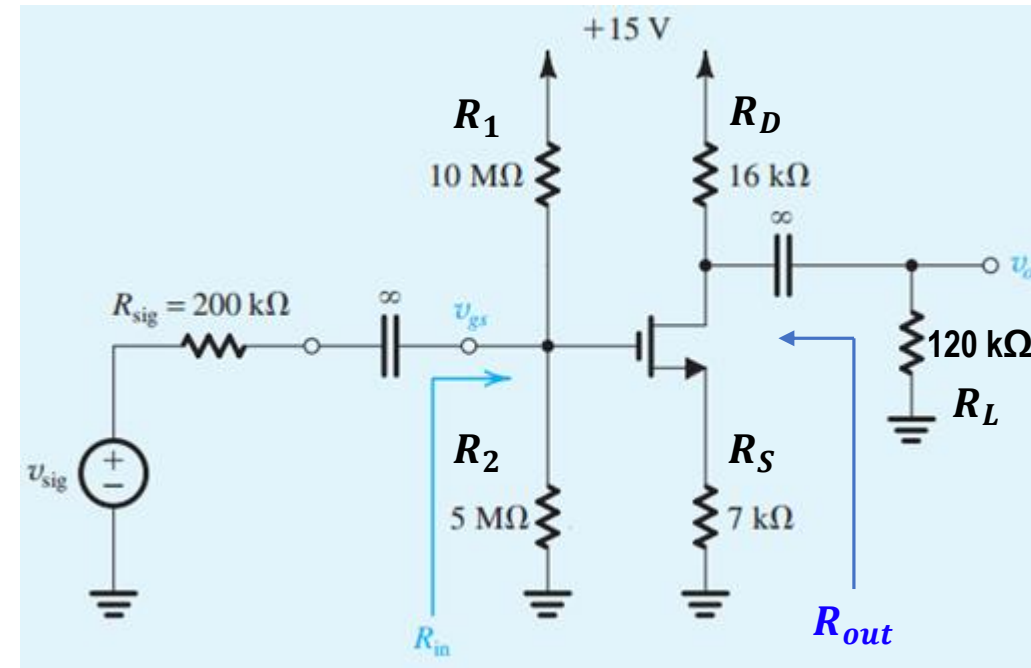
□ The figure shows a discrete-circuit amplifier using an NMOS transistor with $V_{tn} = 1\text{ V}$, $\beta_n = 4\text{ mA/V}^2$, and $\lambda_n = 0.01\text{ V}^{-1}$. Find the following.

□ Q-point (V_{GS} , V_{DS} , and I_D)

□ V_{ov} , g_m and r_o

□ Draw the small-signal equivalent circuit for the amplifier.

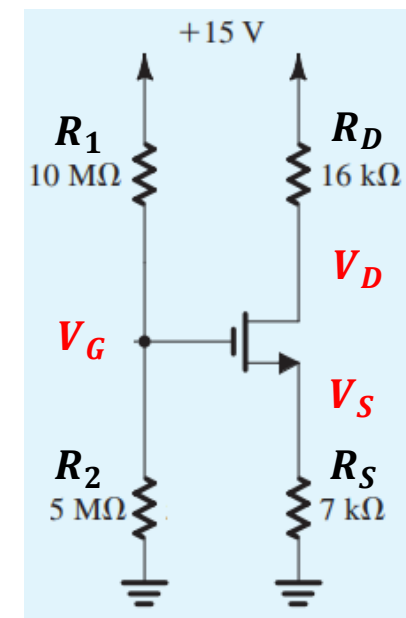
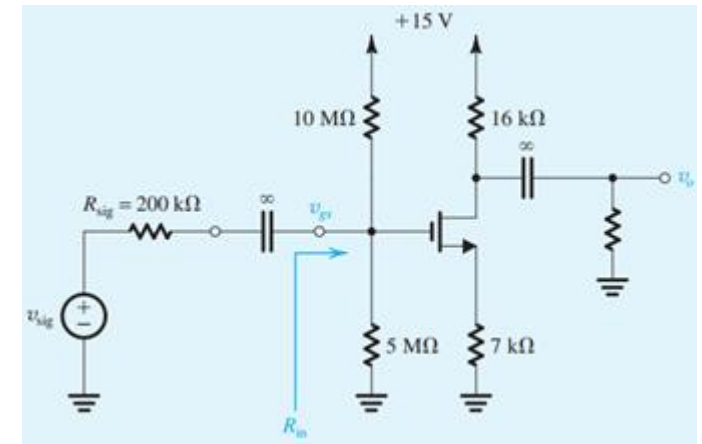
□ Find R_{in} , R_{out} , and $A_v = \frac{v_o}{v_{sig}}$



Example #2 (2/7)

DC Analysis

- ❑ Cancel all AC sources
 - ❑ AC VS (S.C), AC CS (O.C)
- ❑ At DC $\rightarrow f = 0 \text{ Hz}$
 - ❑ Capacitors: $X_C = 1 / 2\pi fC = 1 / 0 = \infty \Omega$ (O.C)
 - ❑ Inductors: $X_L = 2\pi fL = 0 \Omega$ (S.C)
- ❑ Solve input and output loops to get the Q-point. Assume that the transistor is in saturation region and get Q-point. Then check, if not in saturation, repeat analysis in triode region
- ❑ Note: Generally, the channel length modulation effect is ignored in DC analysis for simplicity. ($\lambda = 0$).



Example #2 (3/7)

❑ Assume that the transistor is in saturation region

$$\square I_G = 0, V_G = V_{DD} \frac{R_2}{R_1 + R_2} = 15 \frac{5 \text{ M}\Omega}{5 \text{ M}\Omega + 10 \text{ M}\Omega} = 5 \text{ V}$$

$$\square V_{GS} = V_G - V_S = 5 \text{ V} - I_D R_S \quad (1)$$

$$\square I_D = \frac{\beta_n}{2} (V_{GS} - V_{tn})^2 \quad (2) \quad \text{Sub. (1) into (2)}$$

$$\square I_D = \frac{\beta_n}{2} (5 \text{ V} - I_D R_S - V_{tn})^2 \quad \text{Solve for } I_D$$

$$I_D = \frac{4 \times 10^{-3}}{2} (5 \text{ V} - 7 \times 10^3 \times I_D - 1)^2$$

$$98 \times 10^3 \times I_D^2 - 113 I_D + 0.032 = 0$$

$$I_D = 6.531 \text{ mA}$$

$$I_D = 0.5 \text{ mA}$$

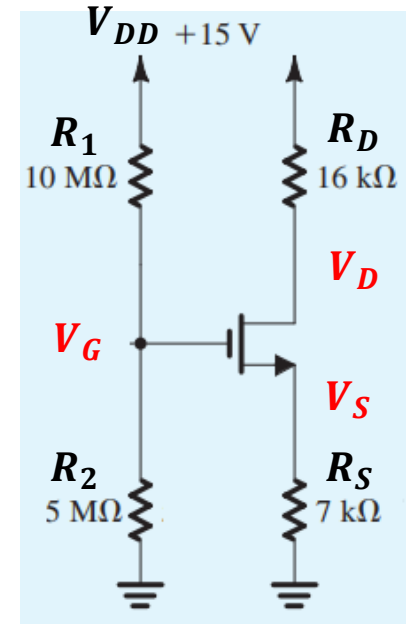
Get V_{GS} from (1)

$$V_{GS} = -40.717 \text{ V}$$

Rejected

$$V_{GS} = 1.5 \text{ V}$$

Accepted



Example #2 (4/7)

□ $I_D = 0.5 \text{ mA}, V_{GS} = 1.5 \text{ V}$

□ $V_D = V_{DD} - I_D \times R_D = 7 \text{ V}$

□ $V_S = I_D \times R_S = 3.5 \text{ V}$

□ $V_{DS} = V_D - V_S = 7 \text{ V} - 3.5 \text{ V} = 3.5 \text{ V}$

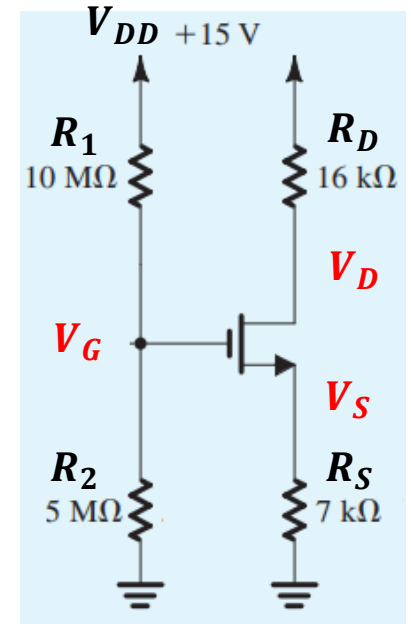
□ Check to be in saturation $\rightarrow V_{DS} > V_{GS} - V_{tn} \rightarrow (3.5 \text{ V}) > (1.5 \text{ V} - 1 \text{ V}) \rightarrow$
OK (saturation)

□ Get V_{ov}, g_m, r_o

□ $V_{ov} = V_{GS} - V_{tn} = 1.5 \text{ V} - 1 \text{ V} = 0.5 \text{ V}$

□ $g_m = \beta_n V_{ov} = (4 \text{ mA/V}^2)(0.5 \text{ V}) = 2 \text{ mS}$

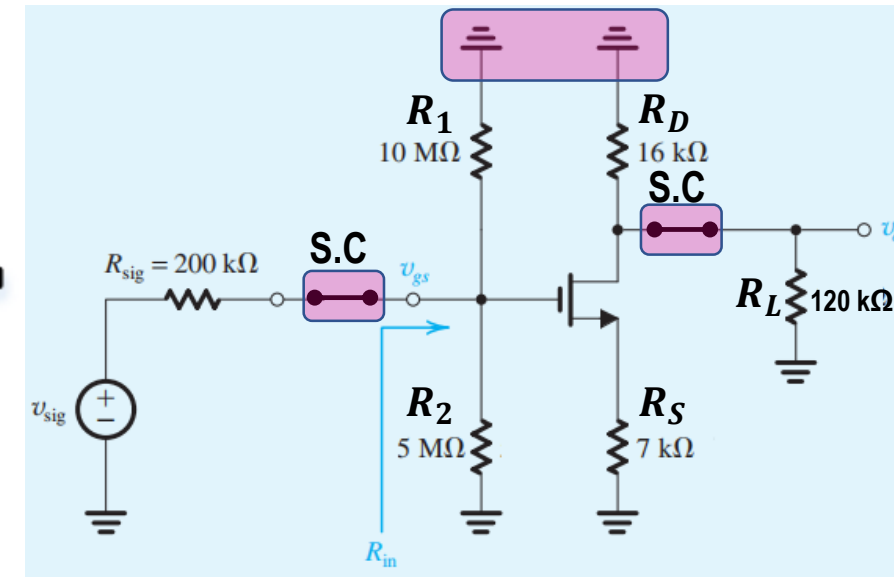
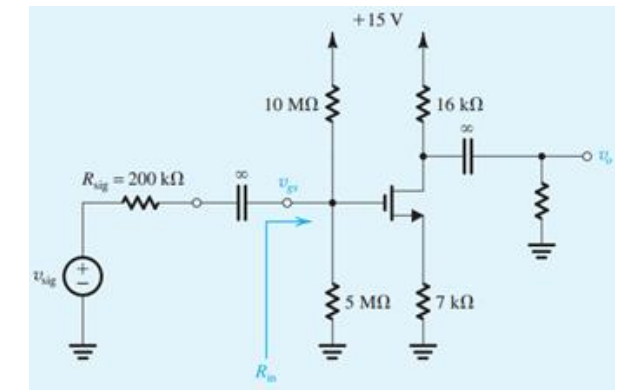
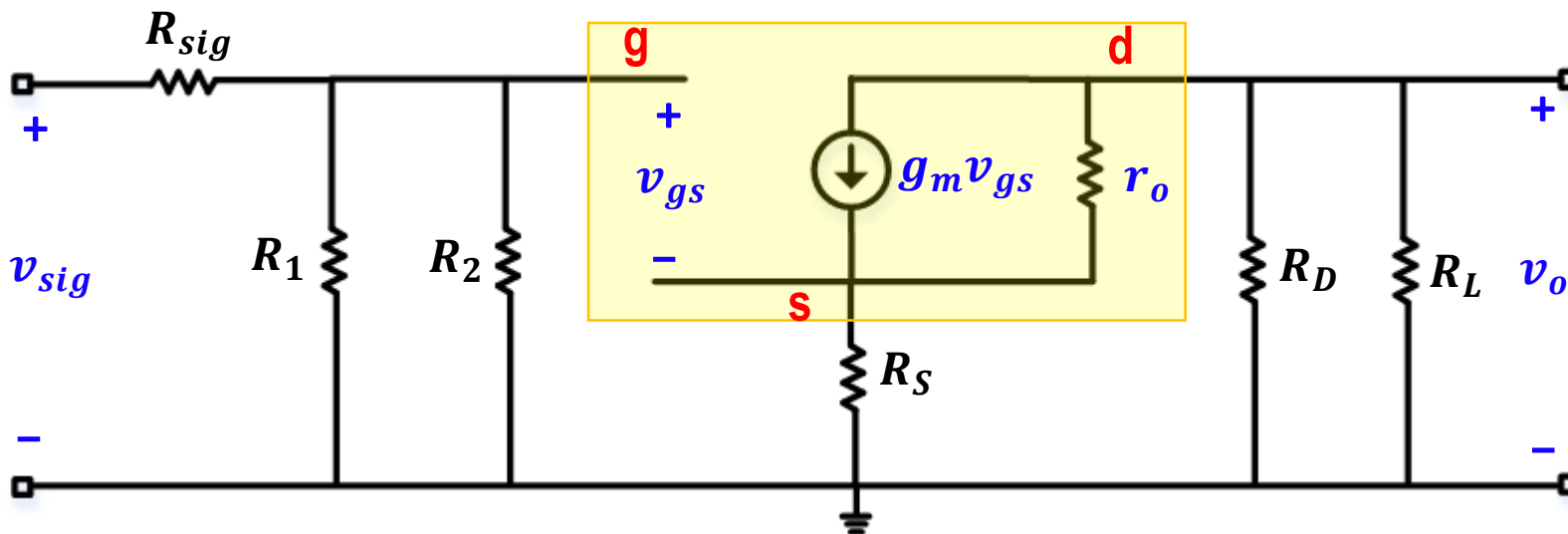
□ $r_o = \frac{1}{\lambda I_D} = \frac{1}{(0.01)(0.5 \text{ mA})} = 200 \text{ K}\Omega$



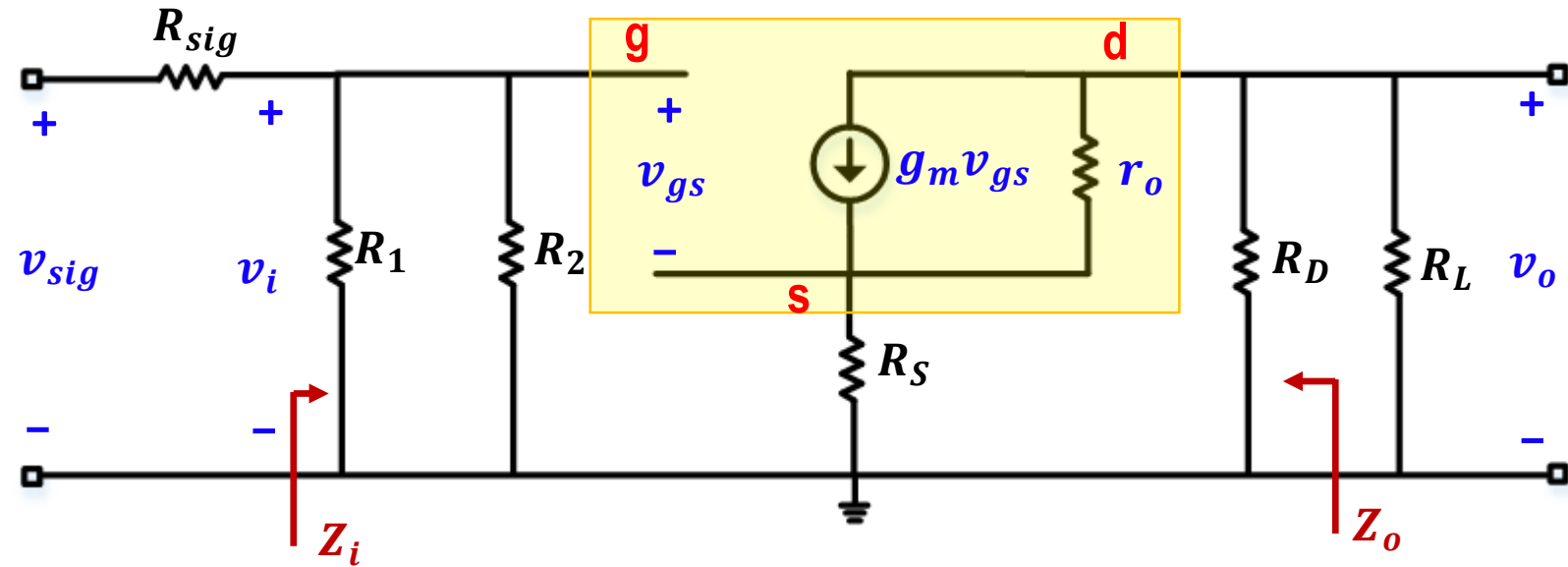
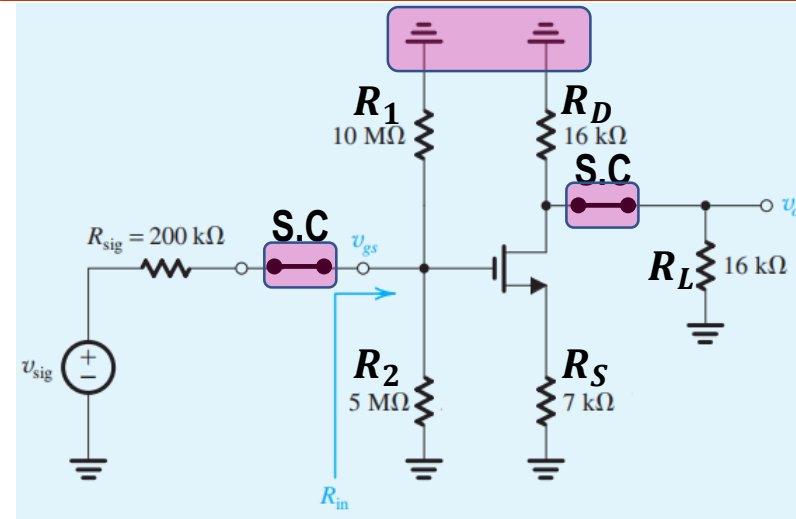
Example #2 (5/7)

AC Analysis

- ❑ Cancel all DC sources $\rightarrow$ DC VS (S.C), DC CS (O.C)
- ❑ At Mid and high band frequencies $\rightarrow f \uparrow \uparrow$
 - ❑ Coupling Capacitors: $X_C = 1 / 2\pi f C \approx 0 \Omega$ (S.C)
- ❑ Draw the small signal equivalent circuit



Example #2 (6/7)



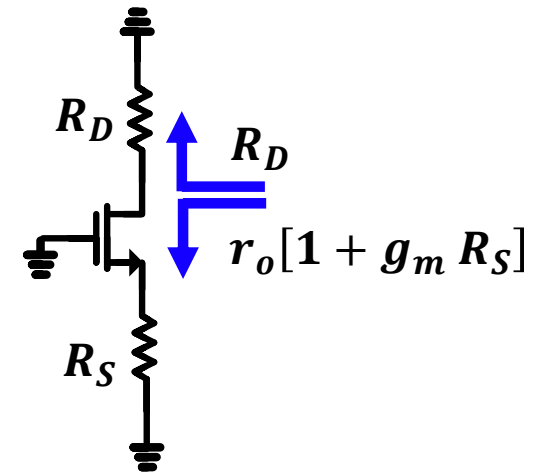
□ $R_{in} = R_1 \parallel R_2 = 10 \text{ M}\Omega \parallel 5 \text{ M}\Omega = 3.33 \text{ M}\Omega$

□ $R_{out} = R_D \parallel r_o[1 + g_m R_S] = R_D \parallel 15 r_o = 16 \text{ K}\Omega \parallel 3 \text{ M}\Omega = 15.92 \text{ K}\Omega$

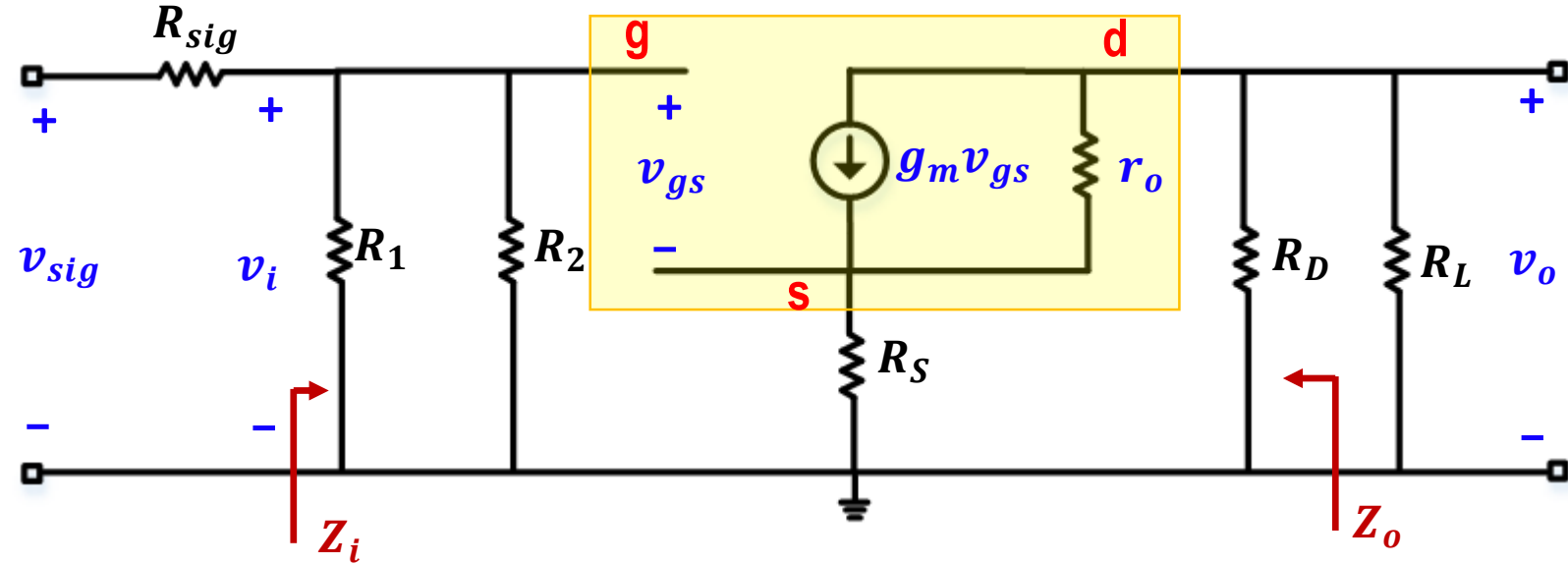
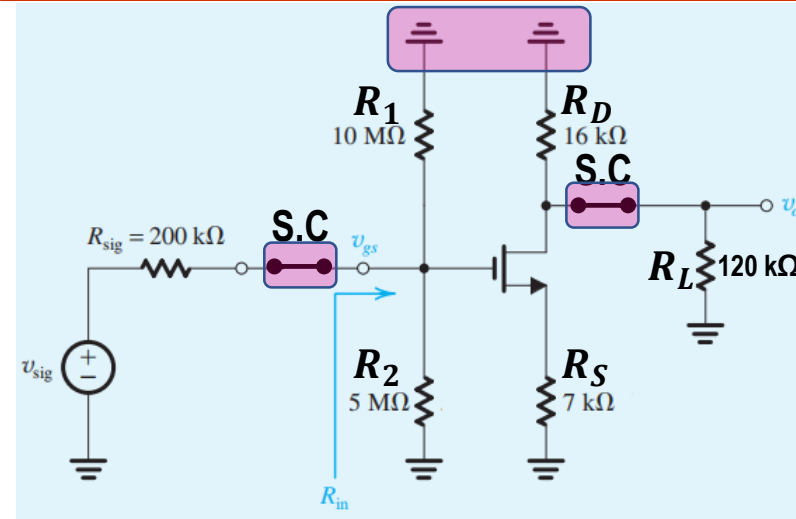
□ Get $A_{vs} = v_o/v_{sig}$ [ignore r_o for simplicity]

□ $v_o = -g_m v_{gs}[R_D \parallel R_L]$

□ $v_i = v_{gs} + g_m v_{gs} R_S \rightarrow v_{gs} = \frac{v_i}{1 + g_m R_S}$



Example #2 (7/7)



$$\square v_o = \frac{-g_m v_i [R_D \parallel R_L]}{1 + g_m R_S} \rightarrow \frac{v_o}{v_i} = \frac{-g_m [R_D \parallel R_L]}{1 + g_m R_S}$$

$$\square v_i = v_{sig} \frac{R_{in}}{R_{in} + R_{sig}} \rightarrow \frac{v_i}{V_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}}$$

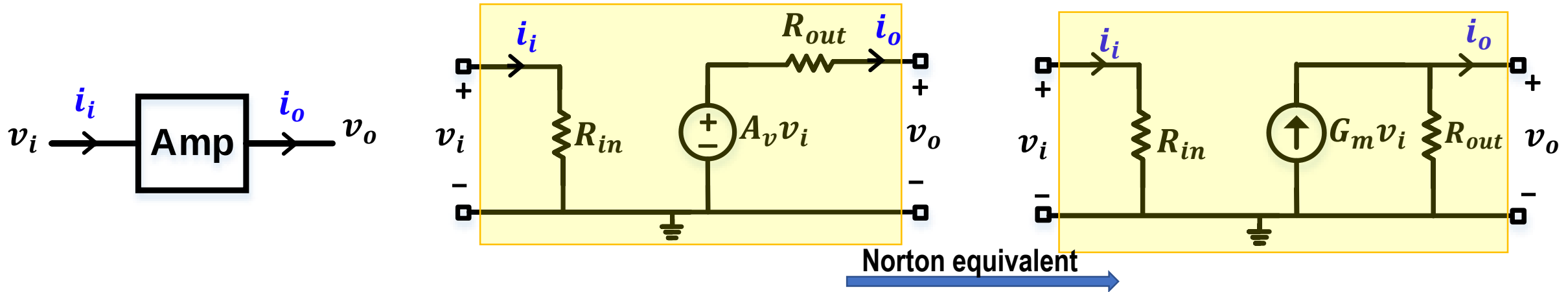
$$\square A_{vs} = \frac{v_o}{V_{sig}} = \frac{v_o}{v_i} \times \frac{v_i}{v_{sig}} = -\frac{R_{in}}{R_{in} + R_{sig}} \times \frac{g_m [R_D \parallel R_L]}{1 + g_m R_S}$$

$$= -0.9434 \times 1.8824 = -1.776 \text{ V/V}$$

Outline

- ☐ Course Information
- ☐ Introduction: Semiconductors between past, present, and future
- ☐ Quick Review
 - ☐ Bipolar Junction Transistor (BJT)
 - ☐ Metal Oxide Semiconductor Field Effect Transistor (MOSFET)
- ☐ Amplifier Model

Amplifier Model



Input and output impedances:

R_{in} , R_{out}

Voltage gain:

$$\frac{v_o}{v_i} = \frac{A_v v_i}{v_i} = A_v$$

Short circuit current gain:

$$\frac{i_{o,sc}}{i_i} = \frac{A_v v_i / R_{out}}{v_i / R_{in}} = \frac{A_v R_{in}}{R_{out}}$$

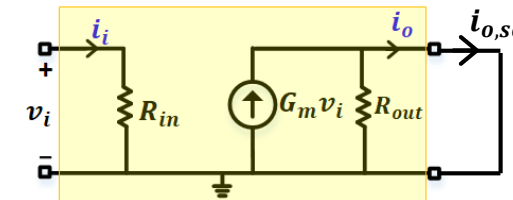
R_{in} , R_{out}

$$\frac{v_o}{v_i} = \frac{G_m v_i R_{out}}{v_i} = G_m R_{out}$$

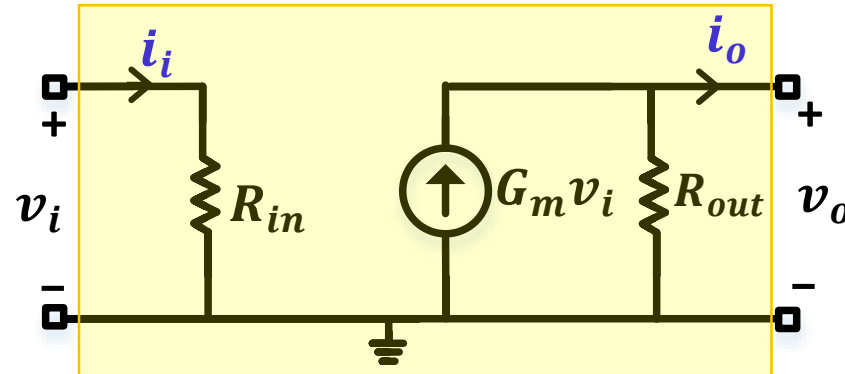
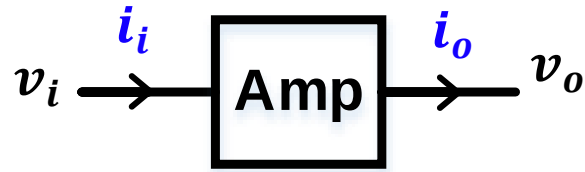
$$\frac{i_{o,sc}}{i_i} = \frac{G_m v_i}{v_i / R_{in}} = G_m R_{in}$$

❑ R_{in} and R_{out} can directly obtained using impedance shortcut

❑ G_m can easily obtained $\rightarrow G_m = \frac{i_{o,sc}}{v_i}$



Amplifier Model



❑ Fast method to perform AC analysis

❑ Get R_{in} and R_{out} using impedance shortcut

❑ Get $G_m = \frac{i_{o,sc}}{v_i}$

❑ Get $A_v = G_m R_{out}$

Common Source Amplifier

□ Find R_{in} , R_{out} , and A_v . Note: this is the equivalent AC circuit and detailed biasing is not shown.

$$\square R_{in} = \infty$$

$$\square R_{out} = R_D \parallel r_o [1 + g_m R_S] \approx R_D$$

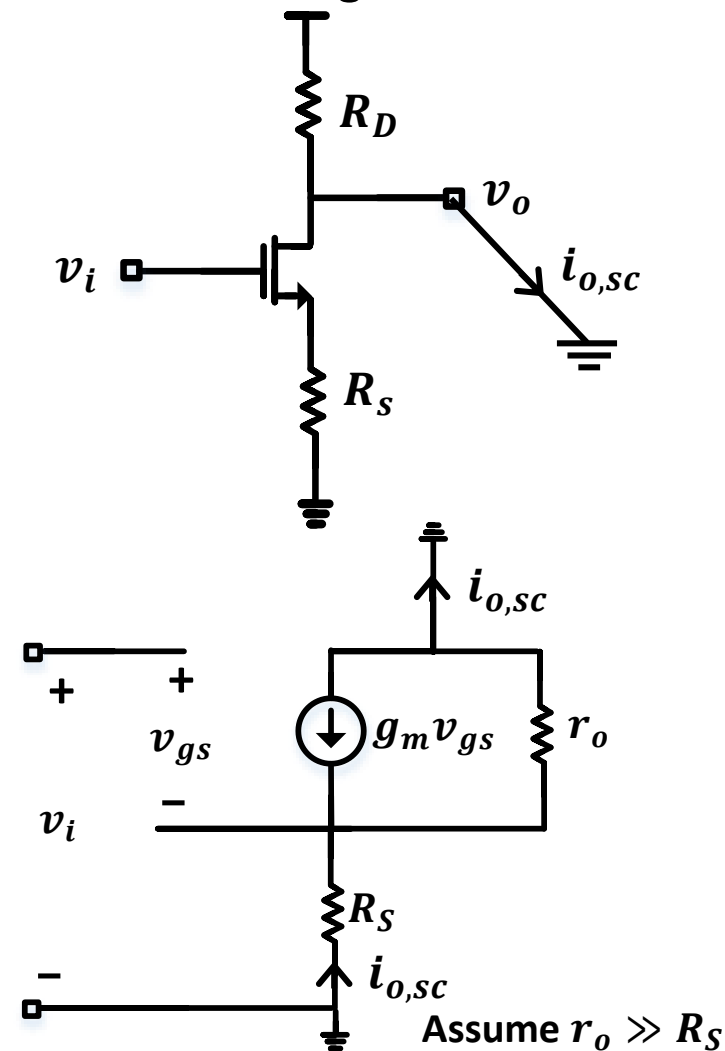
$$\square G_m = \frac{i_{o,sc}}{v_i}$$

$$\square i_{o,sc} = -g_m v_{gs}$$

$$\square v_i = v_{gs} + g_m v_{gs} R_S = v_{gs} [1 + g_m R_S]$$

$$\square G_m = \frac{-g_m}{1 + g_m R_S}$$

$$\square A_v = G_m R_{out} = \frac{-g_m R_D}{1 + g_m R_S}$$



Common Gate Amplifier

□ Find R_{in} , R_{out} , and A_v . Note: this is the equivalent AC circuit and detailed biasing is not shown.

$$\square R_{in} = R_S \parallel \frac{1}{g_m} \left[1 + \frac{R_D}{r_o} \right] \approx R_S \parallel \frac{1}{g_m}$$

$$\square R_{out} = R_D \parallel r_o \approx R_D$$

Assume $r_o \gg R_D$

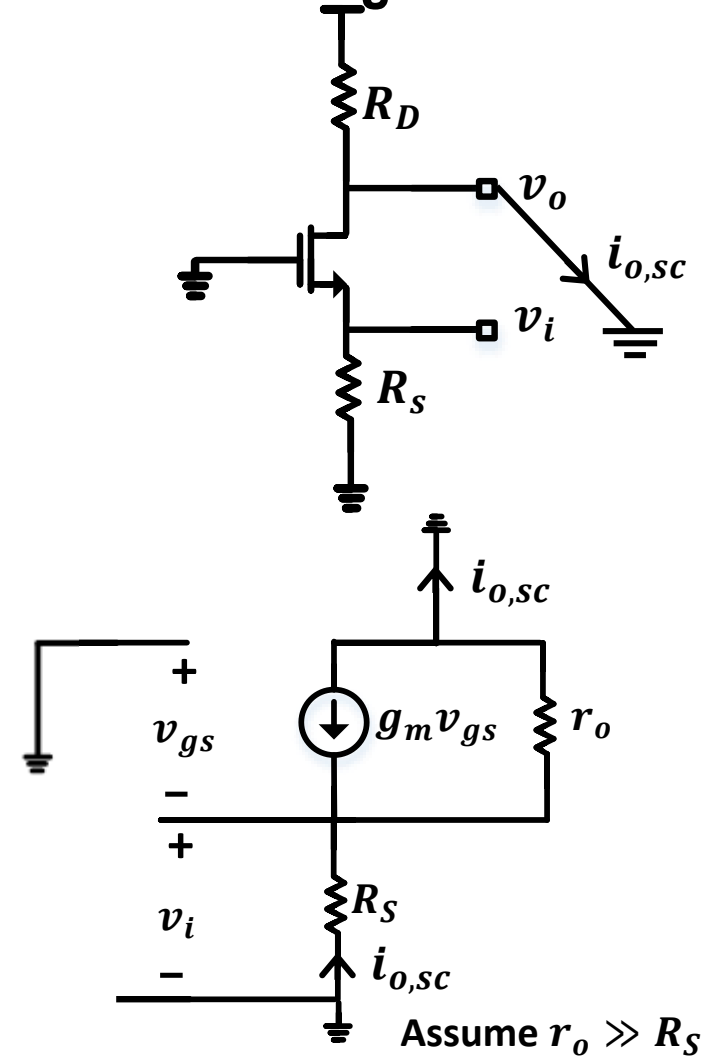
$$\square G_m = \frac{i_{o,sc}}{v_i}$$

$$\square i_{o,sc} = -g_m v_{gs}$$

$$\square v_i = -v_{gs}$$

$$\square G_m = g_m$$

$$\square A_v = G_m R_{out} = g_m R_D$$



Common Drain Amplifier – Source Follower

Find R_{in} , R_{out} , and A_v . Note: this is the equivalent AC circuit and detailed biasing is not shown.

$$R_{in} = \infty$$

$$R_{out} = R_S \parallel \frac{1}{g_m} \left[1 + \frac{R_D}{r_o} \right] \approx R_S \parallel \frac{1}{g_m} = \frac{R_S}{1 + g_m R_S}$$

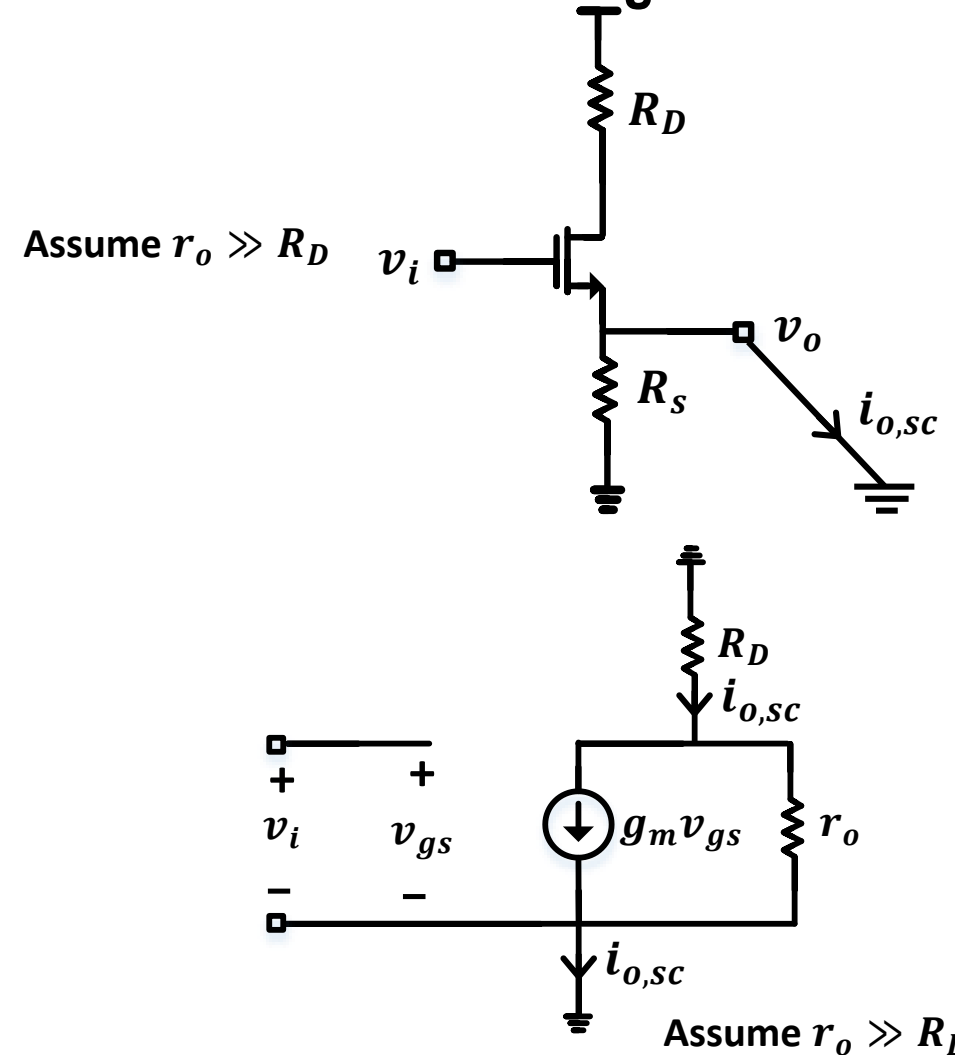
$$G_m = \frac{i_{o,sc}}{v_i}$$

$$i_{o,sc} = g_m v_{gs}$$

$$v_i = v_{gs}$$

$$G_m = g_m$$

$$A_v = G_m R_{out} = \frac{g_m R_S}{1 + g_m R_S} < 1$$



References

- Robert L. Boylestad and Louis Nashelsky, “**Electronic Devices and Circuit Theory**”, 11th edition, Pearson, 2014.
- Adel S. Sedra and Kenneth C. Smith, “**Microelectronic Circuits**”, 7th edition, Oxford University Press, 2016.
- Behzad Razavi, “**Fundamentals of Microelectronics**”, 2th edition, Wiley, 2014.

□ Thanks